



Mobility Behavior Representation & Clustering

Cyrus Shahabi, Ph.D.

*Helen N. and Emmett H. Jones Professor of Engineering
Professor of Computer Science, Electrical & Computer Engineering, and
Spatial Sciences*

Director, Integrated Media Systems Center (IMSC)

*Viterbi School of Engineering
University of Southern California*

Los Angeles, CA 90089-0781

shahabi@usc.edu

OUTLINE



- Mobility Behavior Clustering
 - DETECT
 - VAMBC
- Synthetic Trajectory Generation
 - CSGAN
 - MPGAIL

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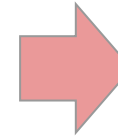
Motivation



Mobility behavior:
the travel activity that describes a user's movements, e.g., work commute, shopping, school commute, dining



Mobility Behavior



COVID 19



User Profiling



Recommendations



Ads targeting



Insurance

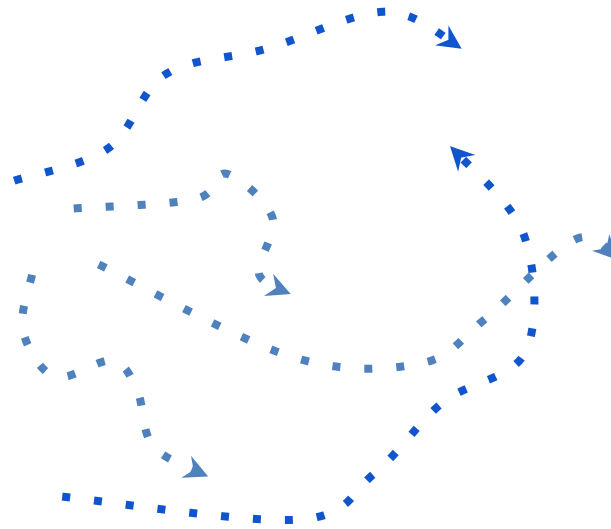
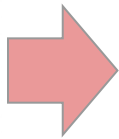


Threats Detection

Idea: Trajectory Data $\rightarrow$ Mobility Behavior



Trajectories

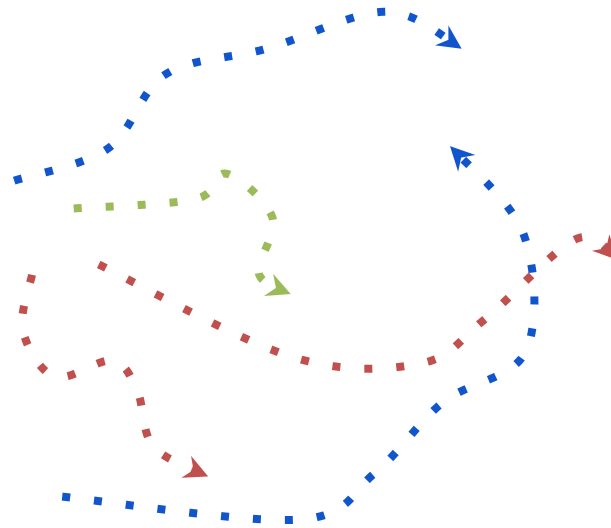
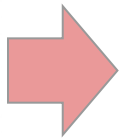


Mobility Behavior

Idea: Trajectory Data → Mobility Behavior



Trajectories



Clusters



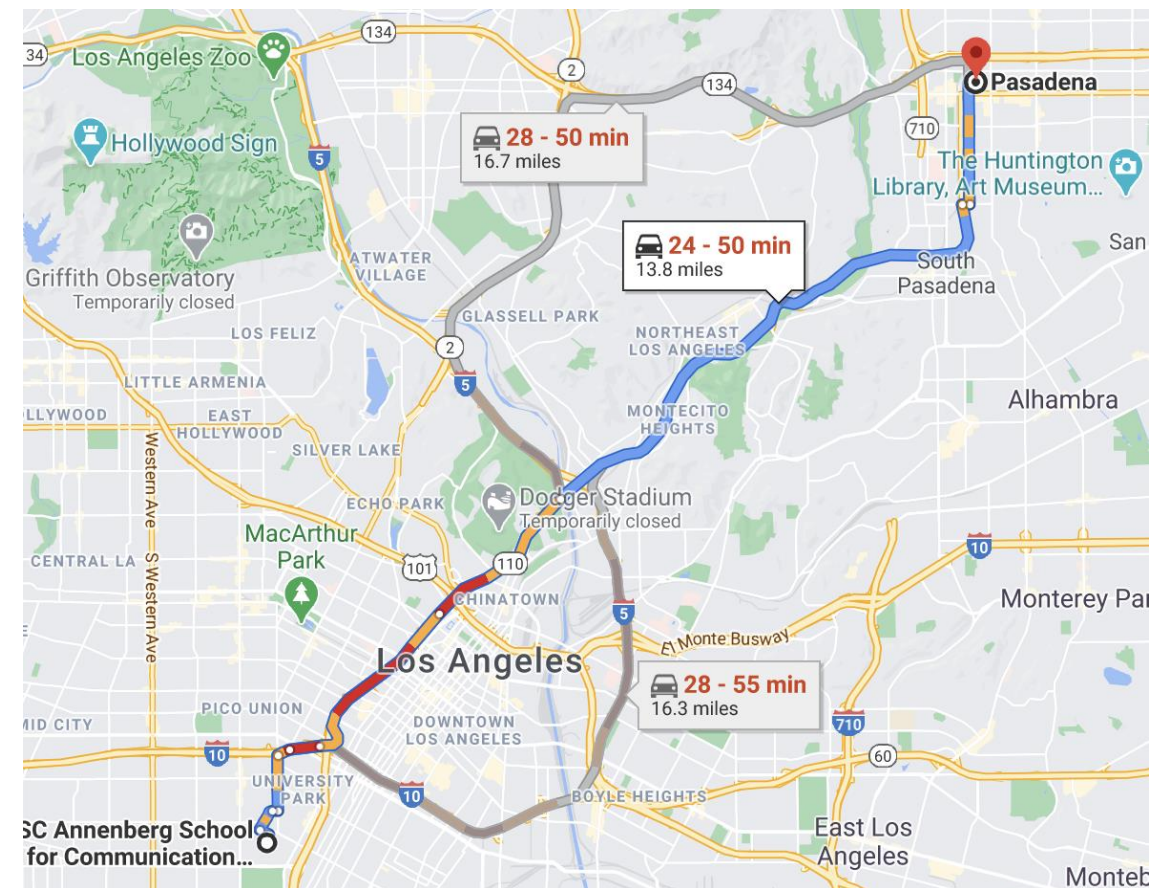
Mobility Behavior



Trajectory Clustering Techniques

- Raw spatio temporal features [**AIR'17**]
- Sequence distance measurement
 - Dynamic Time Warping (**DTW**), Longest Common SubSequence (**LCSS**), Symmetrized Segment-Path Distance (**SSPD**)[**ITS'16**]
- Clustering based on the distances
 - kMeans-DBA[**ICDM'14**], DBSCAN[**CVPR'09**], Hierarchical Clustering

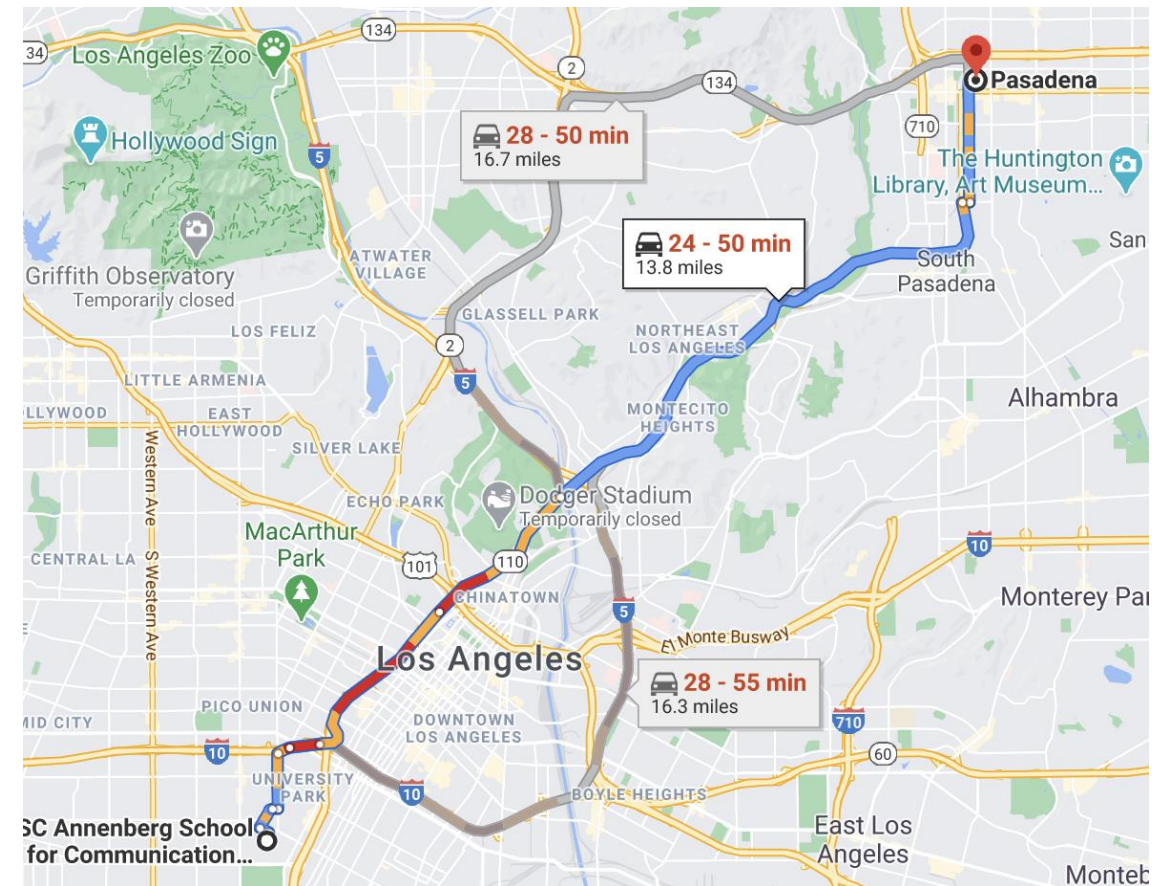
Challenge: Multi-scale Trajectories



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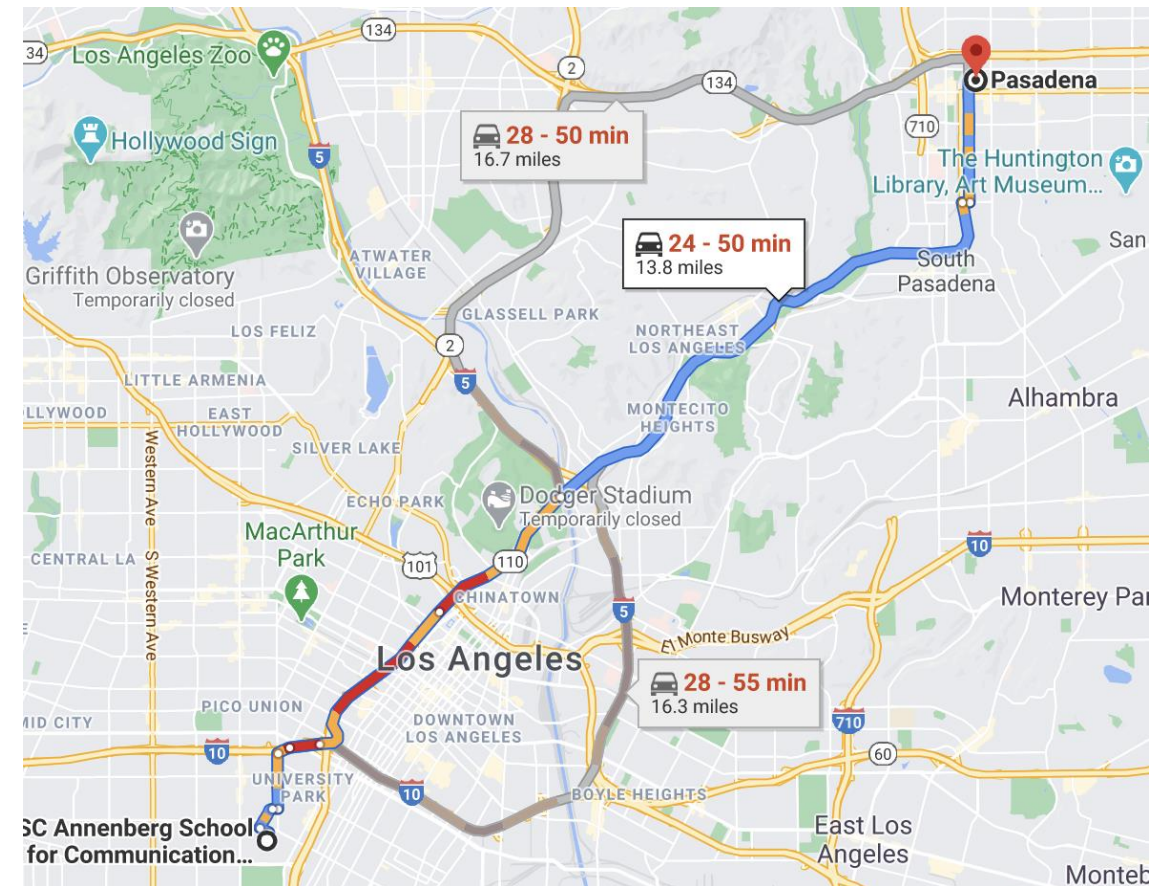
- Different temporal and spatial scales may represent the same mobility behavior



Challenge: Multi-scale Trajectories



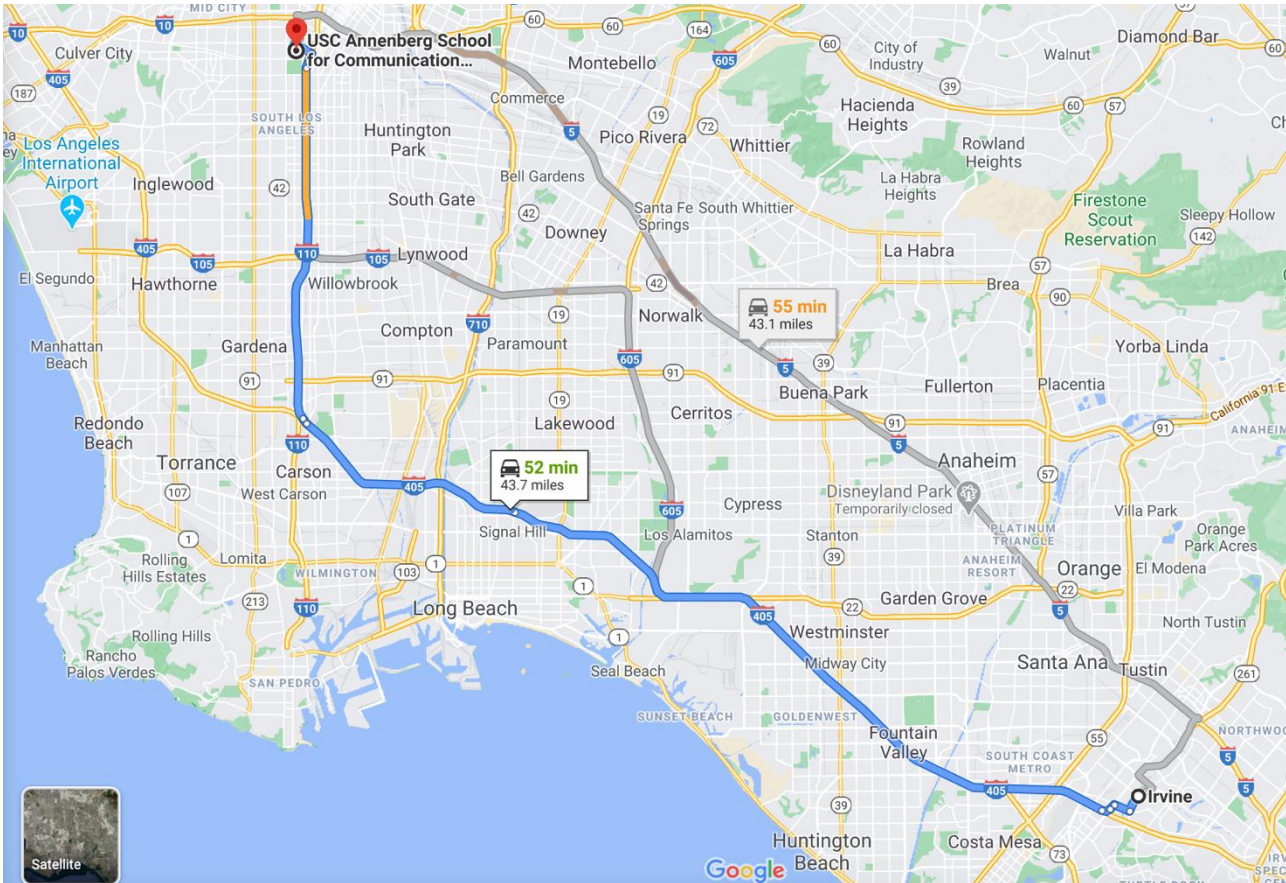
- Different temporal and spatial scales may represent the same mobility behavior
- 50 minutes work commute:
 - 14 miles, 44 miles, 8 miles



Challenge: Multi-scale Trajectories



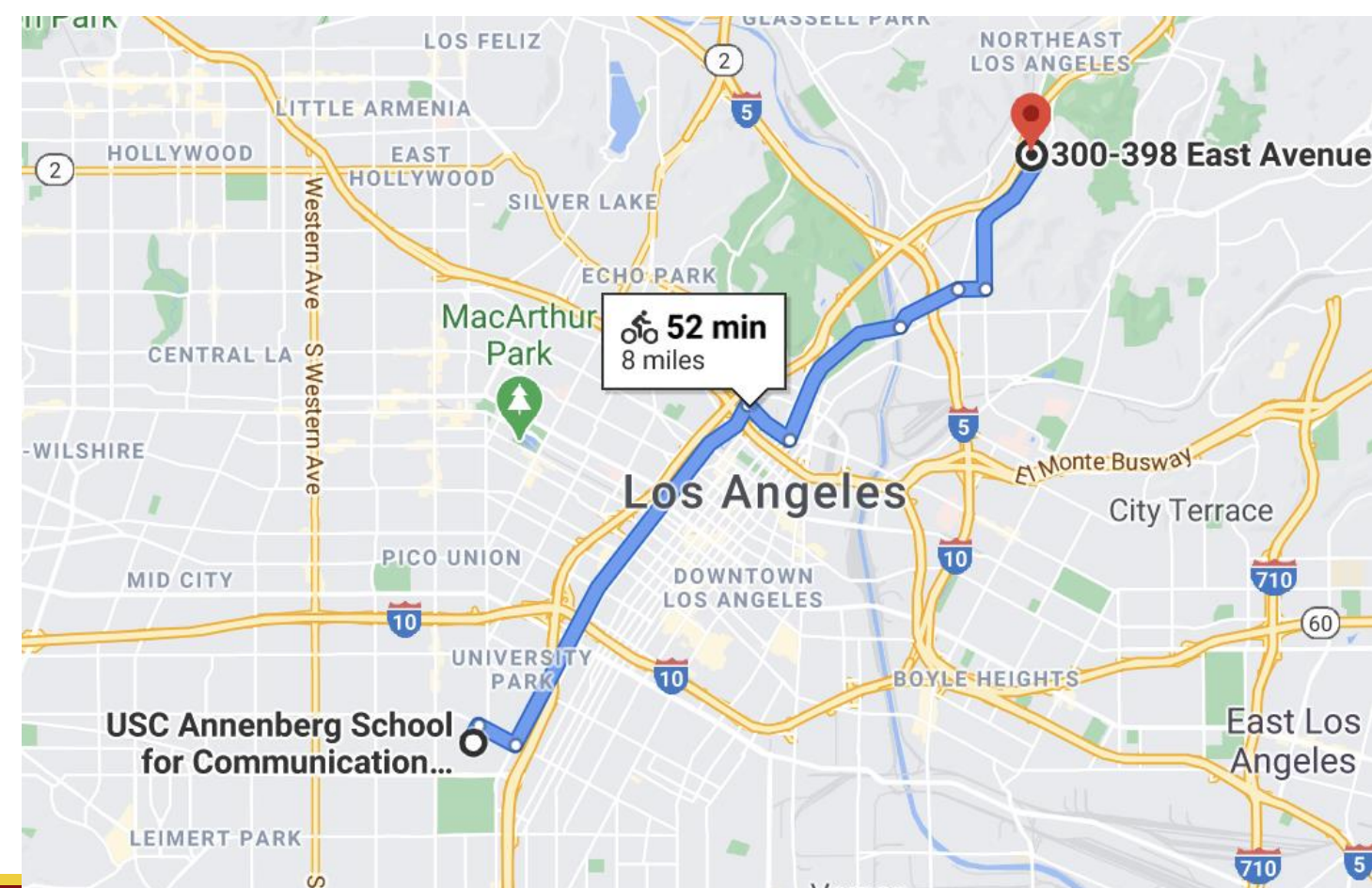
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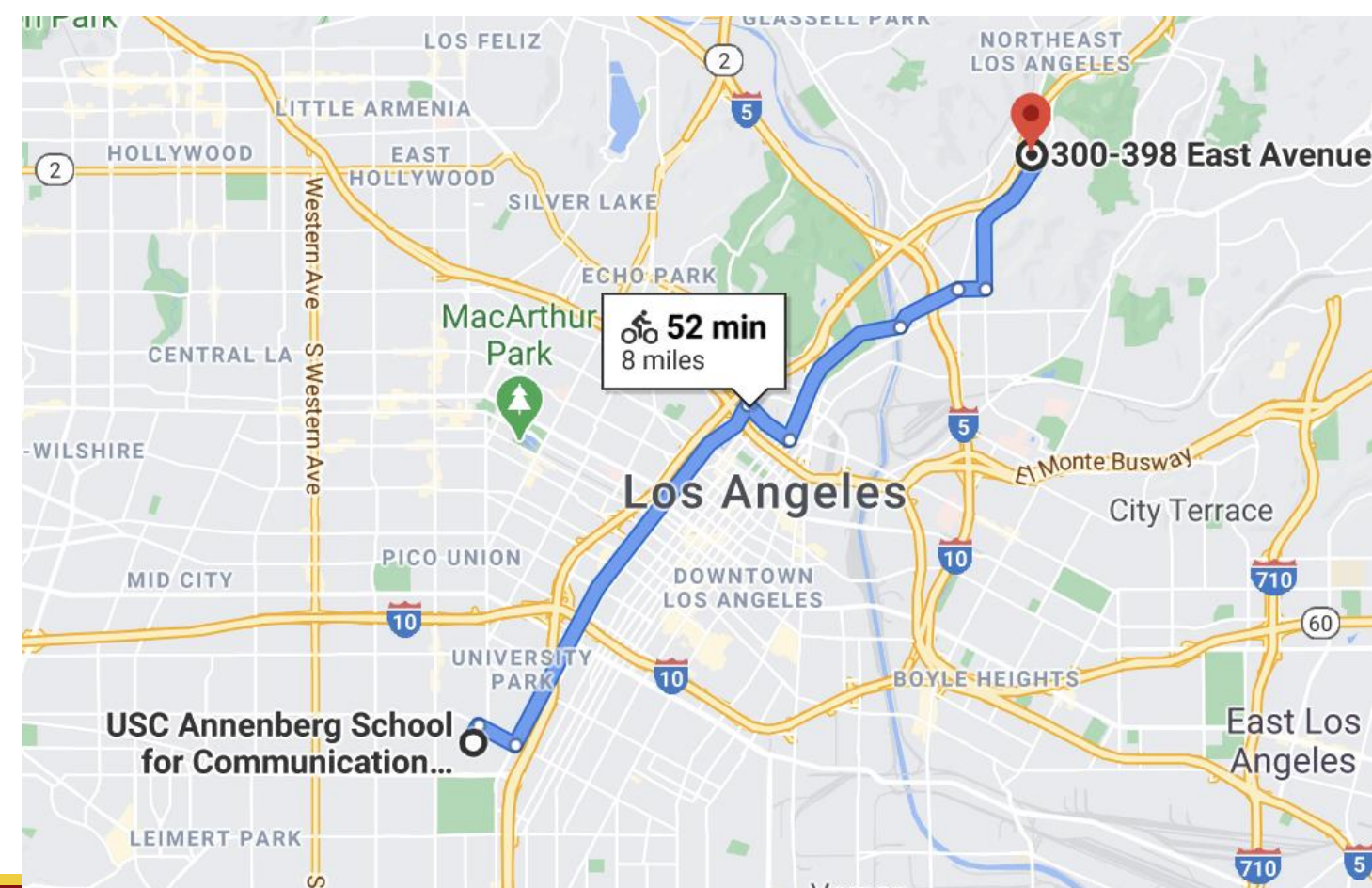
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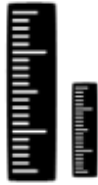


Challenge: Multi-scale Trajectories



- Different temporal and spatial scales may represent the same mobility behavior
- 50 minutes work commute:
 - 14 miles, 44 miles, 8 miles
- 14 miles work commute
 - 20 min, 50 min, 1.5 hour

Limitations of Traditional Trajectory Clustering



Prone to scales & noises 



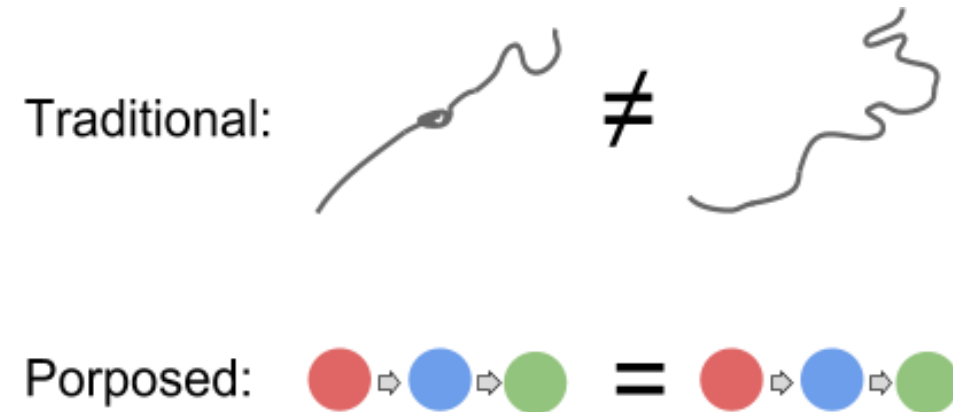
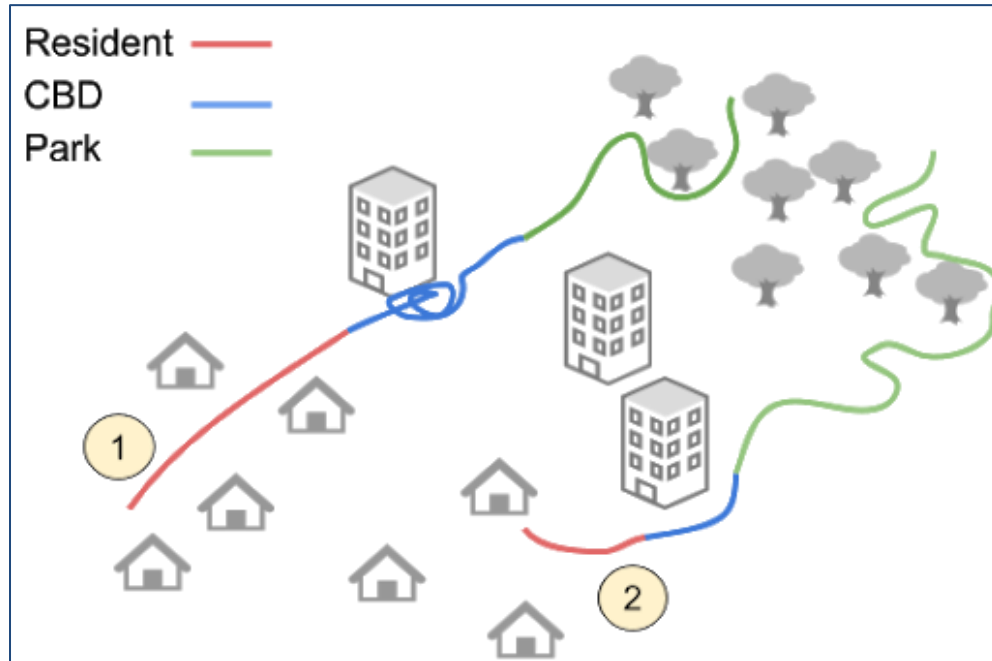
No activity context information 



Pre-defined similarity vs. data-driven 



Intuition



From trajectories to sequences of contexts

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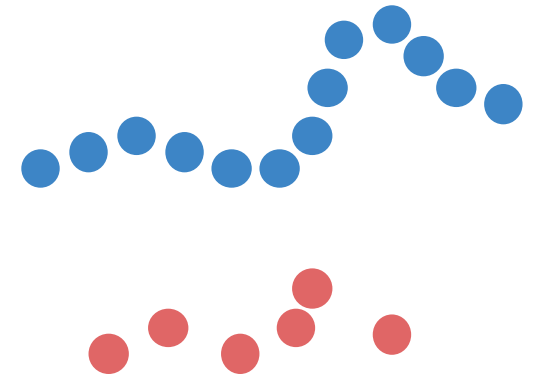
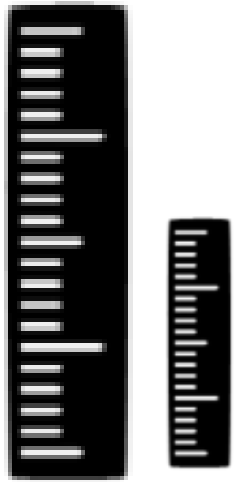


- Approach: DETECT
 - Convert trajectories to sequences of contexts
 - Compact fixed-size representation with RNN
 - Clustering with RNN
- Experiments

Approach



DETECT [BigData 19]



All-scale 

Context-aware 

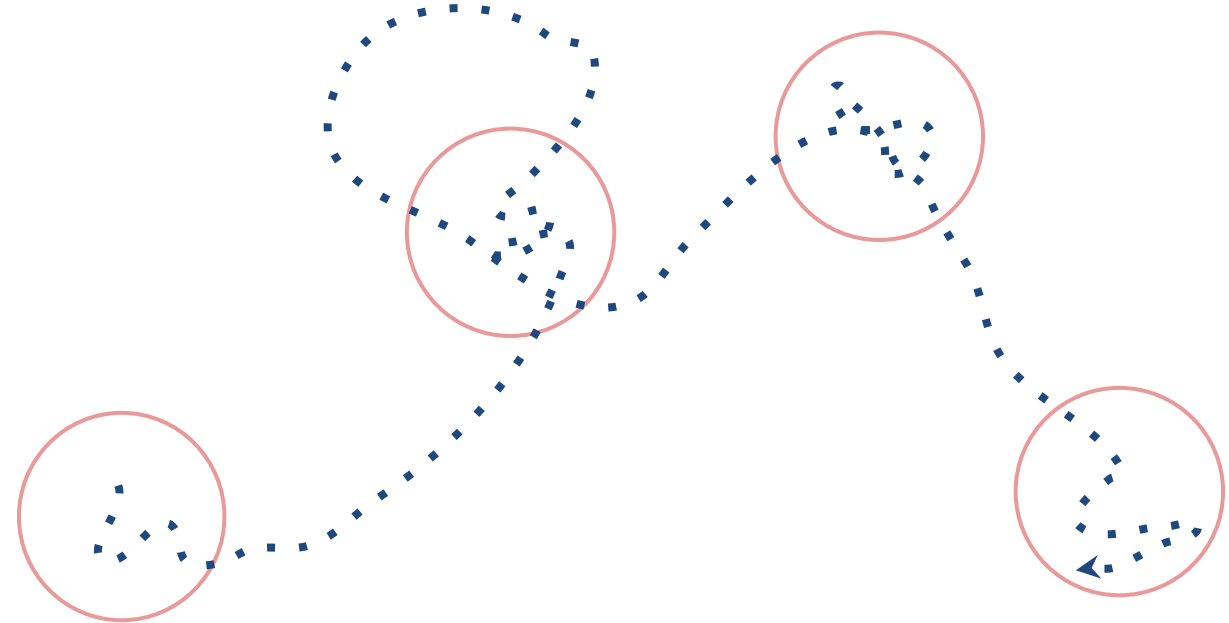
Sequence Dynamics 



All-scale: Stay Point Extraction

Stay points [SIGSPATIAL'08] are representative points that:

1. the user travels within a range of space
2. the user stays in this range for some time

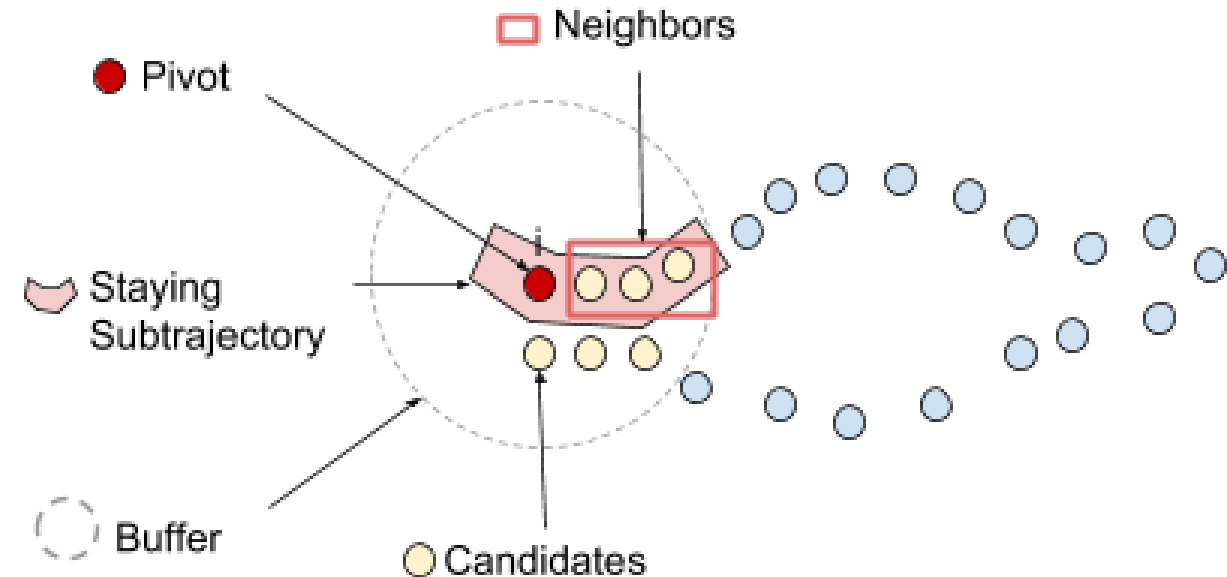




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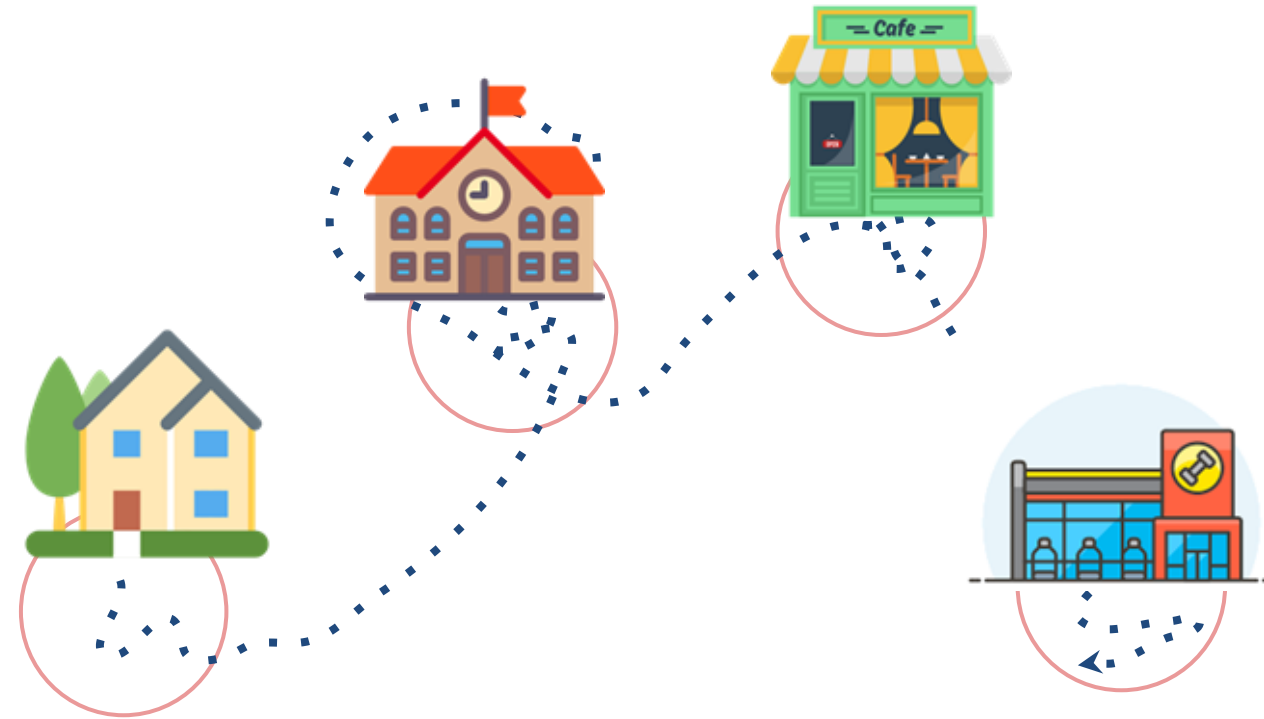




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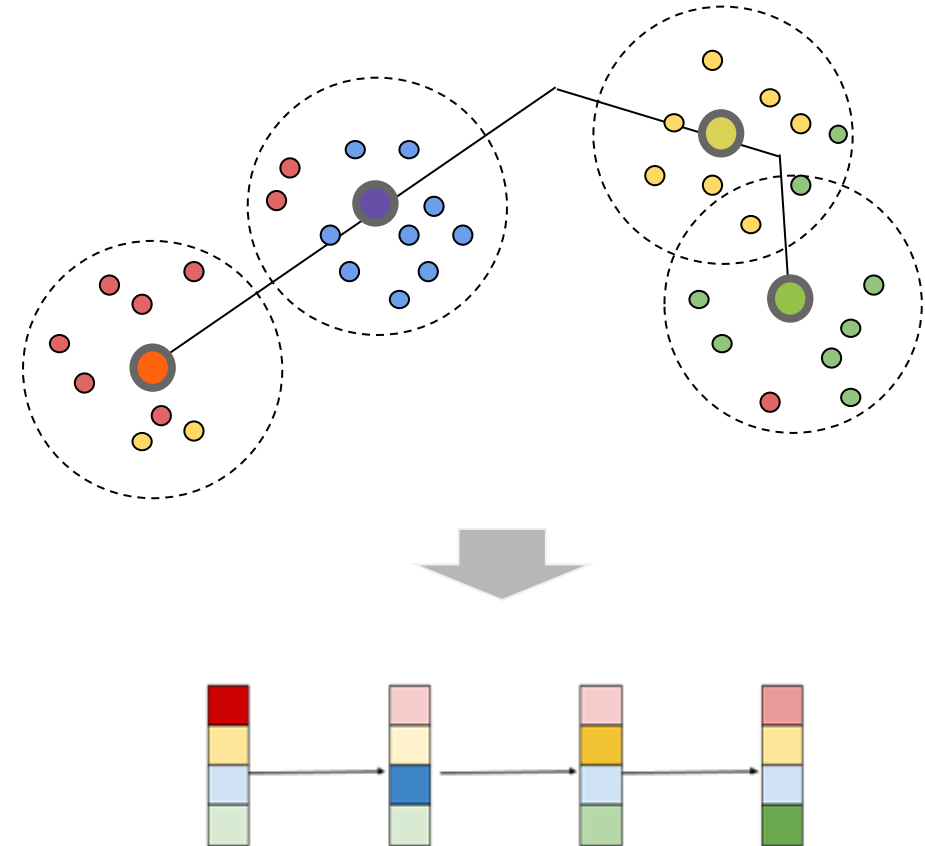
Context-aware: Geographical Augmentation

For each extracted stay point $\dot{s}^{(t)}$:

1. create a spatial buffer $b(r_{poi}, \dot{s}^{(t)})$
2. Search a gazetteer for POI's in the buffer
3. count POIs in the buffer
4. generate a normalized vector

$$x^{(t)} = \{0.3, 0.09, \dots 0.55\}$$

Normalized number of POI categories,
e.g. business area



OUTLINE

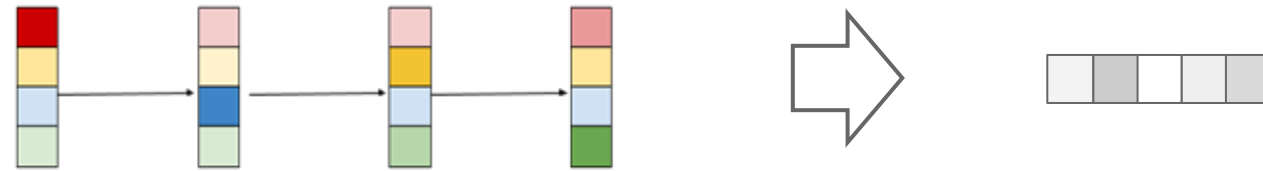


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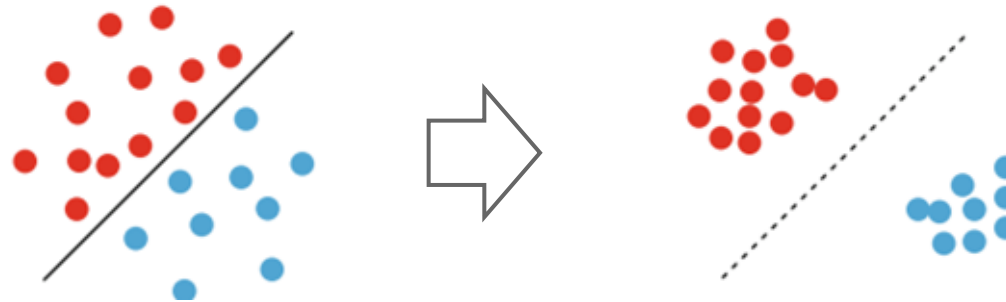


Sequence Dynamics: RNN-AE + Clustering

Phase I:

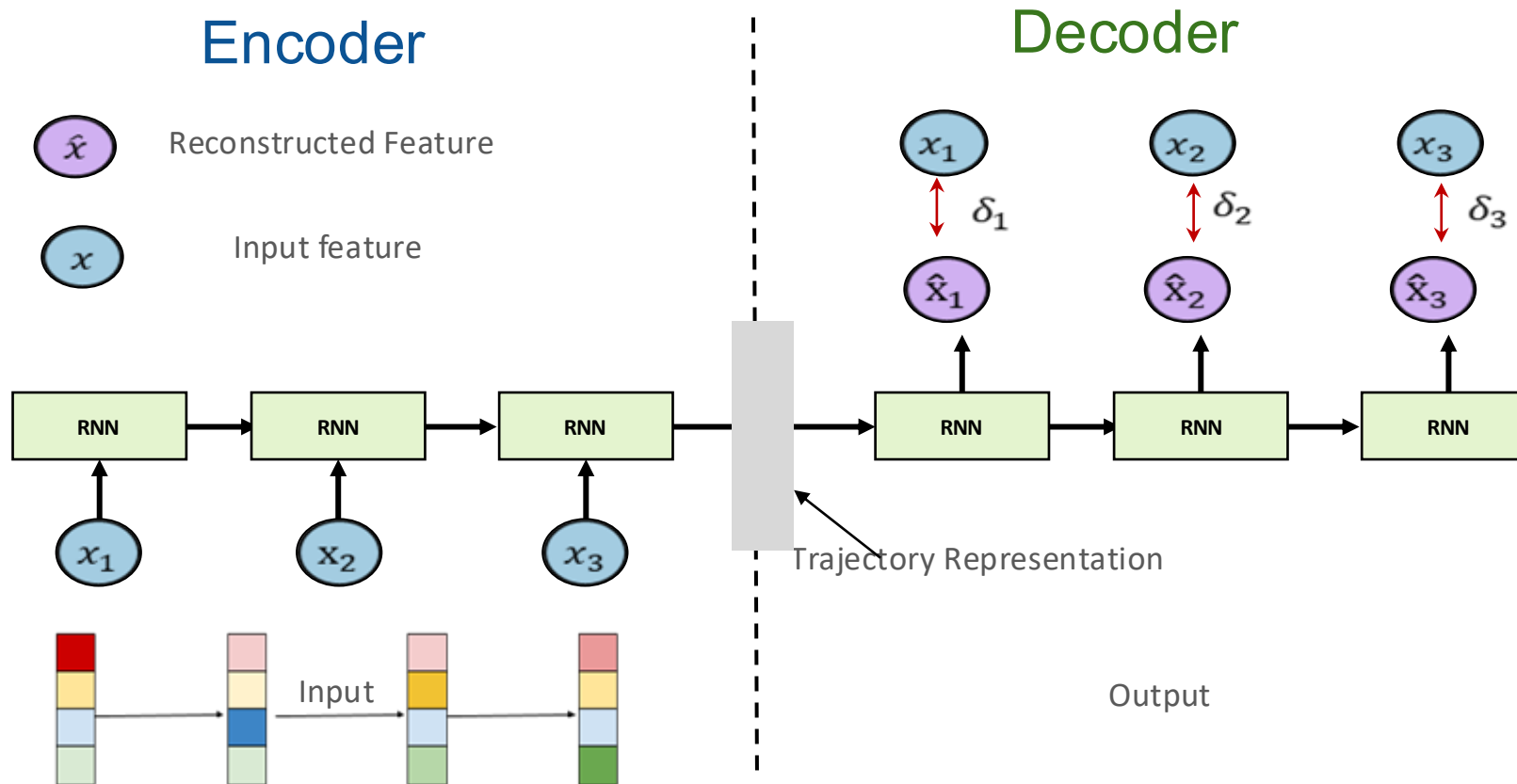


Phase II:





Phase I: RNN Autoencoder

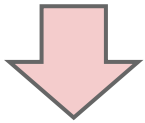


Intuition: Last hidden states of RNN $\rightarrow$ Sequence dynamics

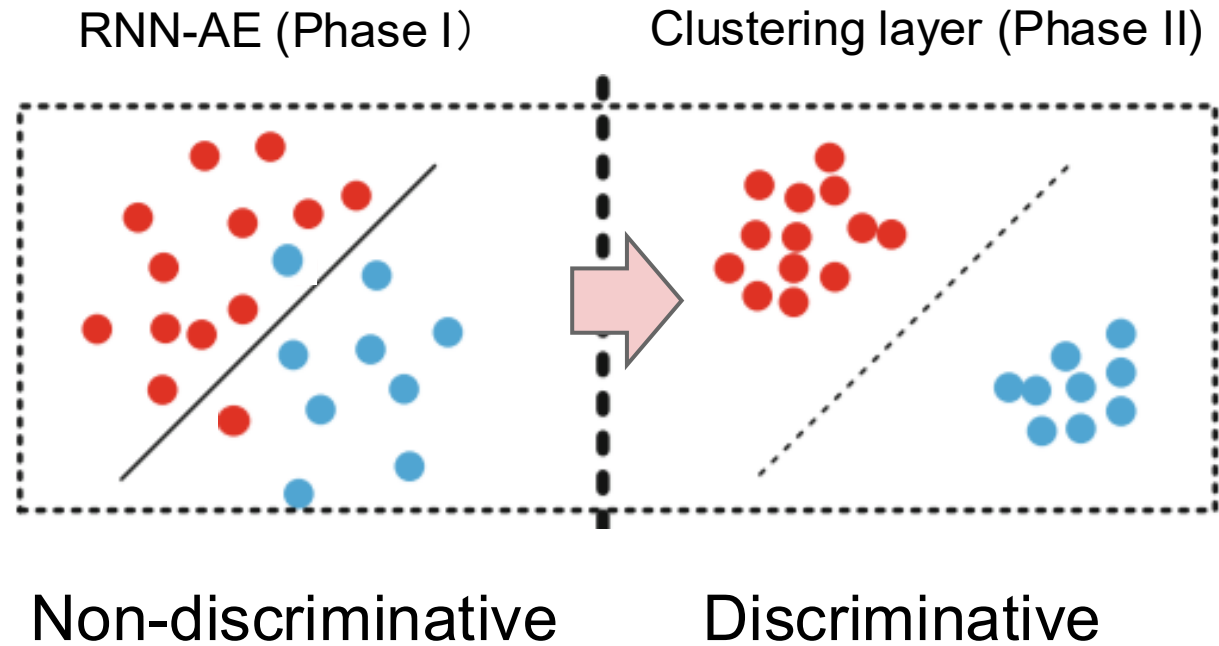


Phase II: Refine for clean clusters

Reconstruction

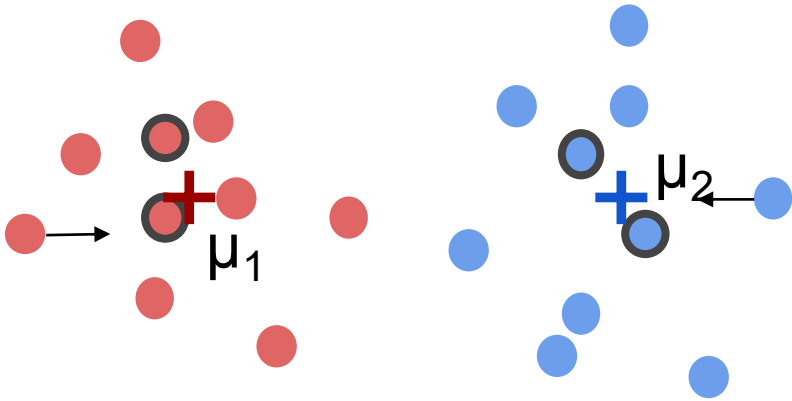


Cluster-aware



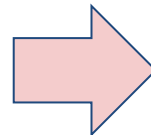


Phase II: Cluster layer

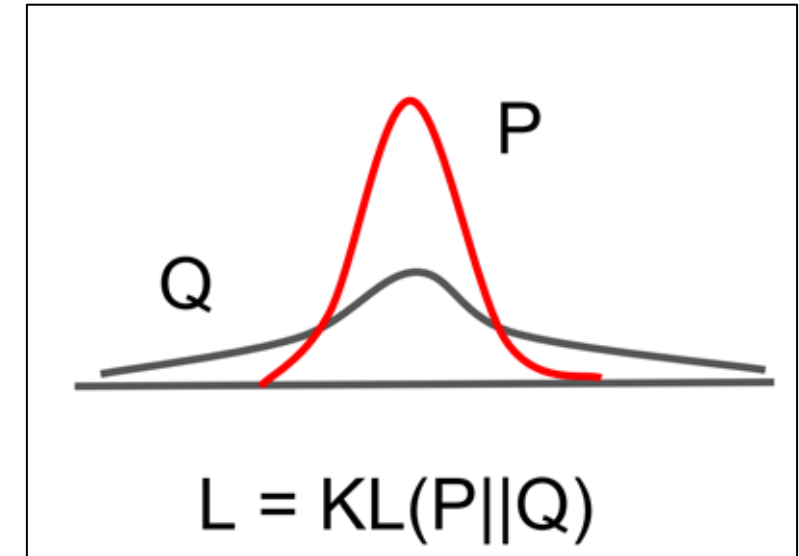


Auxiliary Distribution $P(q)$: hardened probability, trust high-confidence points

$$q_{ij} = \frac{(1 + \|z_i - \mu_j\|^2)^{-1}}{\sum_{j'} (1 + \|z_i - \mu_{j'}\|^2)^{-1}}$$



$$p_{ij} = \frac{q_{ij}^2 / \sum_{i'} q_{i'j}}{\sum_{j'} (q_{ij'}^2 / \sum_{i'} q_{i'j'})}$$





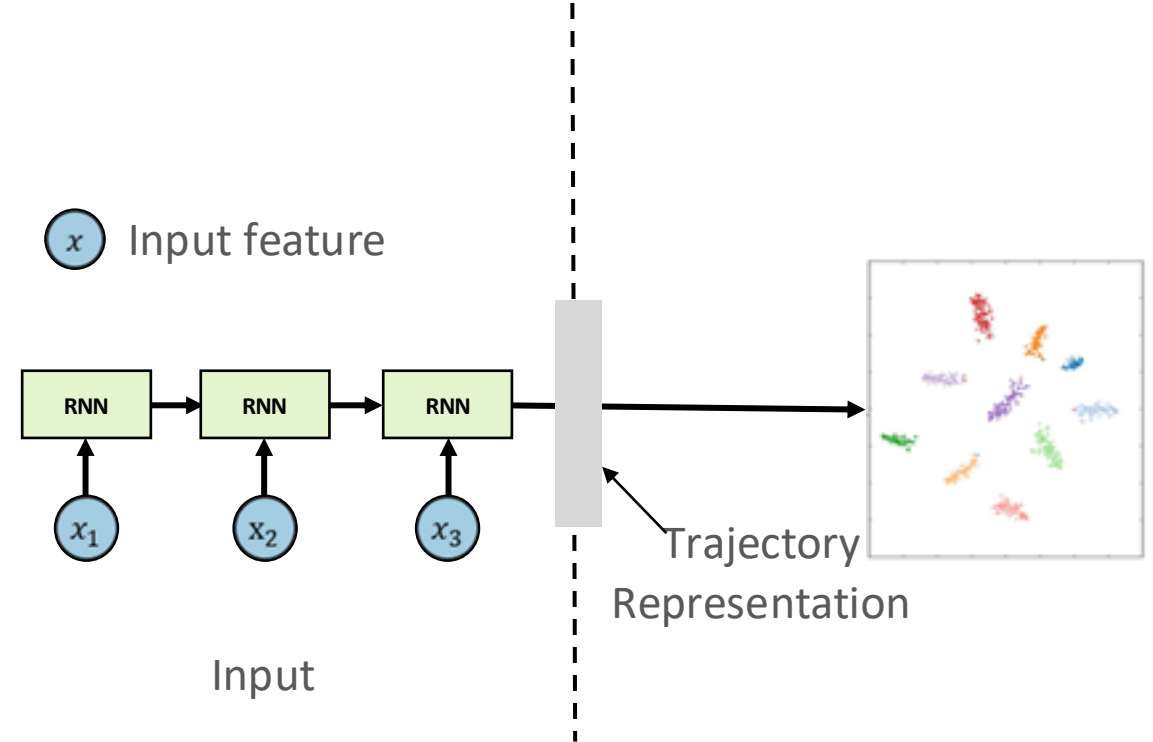
Phase II: Unsupervised optimization

Repeat:

1. Update z (encoder) and μ based on loss:

$$\ell = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

1. After a few batches, update Q and P . Stop if the assignment converges.



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Experimental settings

- Dataset: GeoLife
 - 17,621 trajectories (601 labeled).
 - 6 labels: “dining activities”, “working commutes”, etc.
 - 14,000 POIs in Beijing
- Evaluation Metrics
 - With label: Rand Index (RI), Mutual Information (MI), Purity Fowlkes–Mallows Index (FMI)
 - Without label: Silhouette Score, Dunn index, Within-like Criterion, Between-like Criterion



With-label: quantitative results

Distance **Clustering**

DTW K-Means

LCSS + DBSCAN

SSPD Hierarchical
clustering

Method	RI	MI	Purity	FMI
KM-DBA	0.33	0.64	0.58	0.58
DB-LCSS	0.22	0.55	0.51	0.56
RNN-AE	0.39	0.46	0.56	0.53
SSPD-HCA	0.52	0.93	0.66	0.67
KM-DBA*	0.51	0.91	0.74	0.63
DB-LCSS*	0.5	0.95	0.64	0.66
DETECT Phase I	0.65	1.06	0.84	0.73
DETECT	0.76	1.26	0.89	0.81



With-label: quantitative results



Raw trajectories

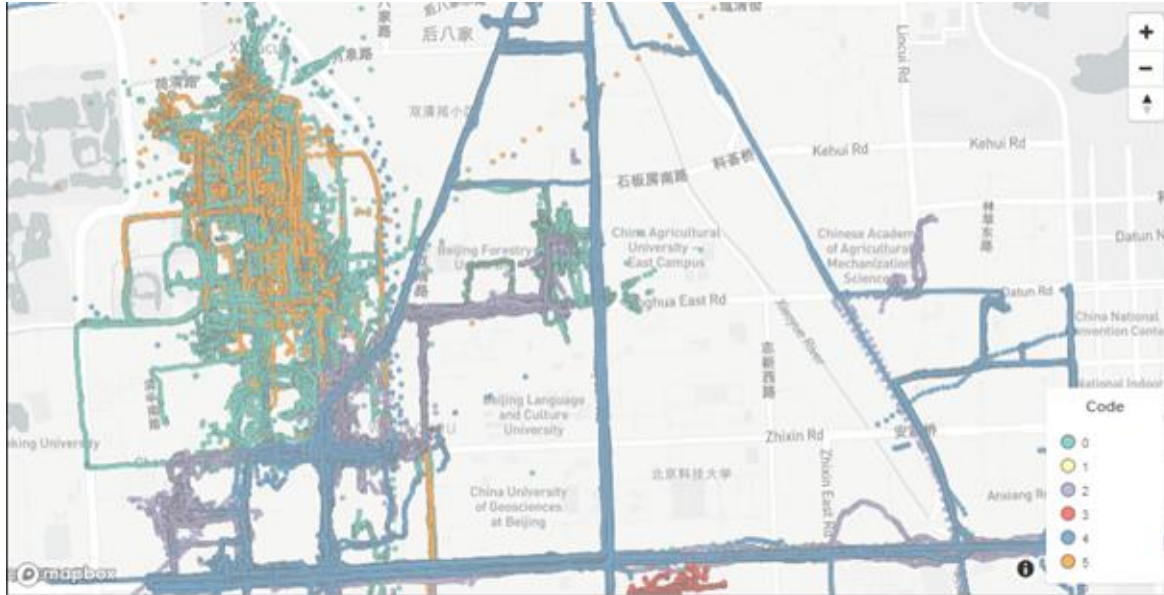
Augmented trajectories



Method	RI	MI	Purity	FMI
KM-DBA	0.33	0.64	0.58	0.58
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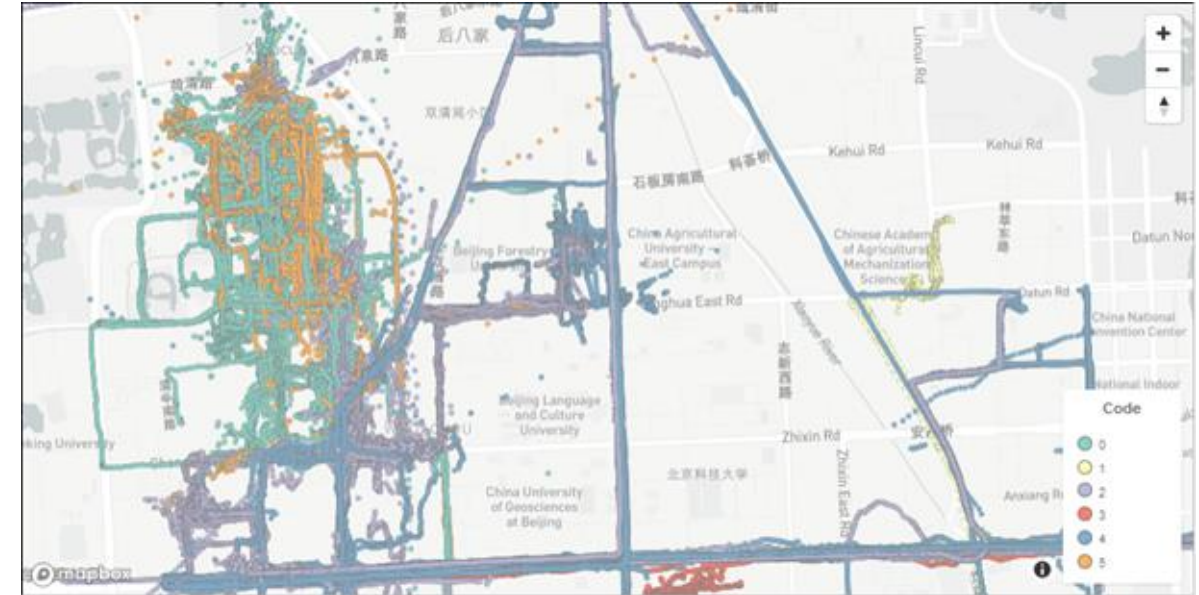


With-label: qualitative results



Ground Truth

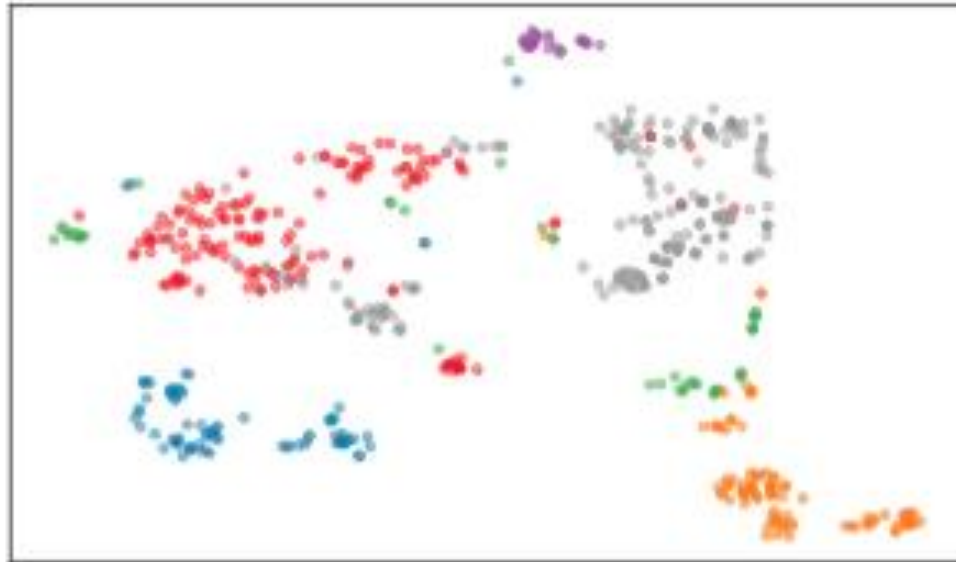
Note: Different colors indicate different clusters.



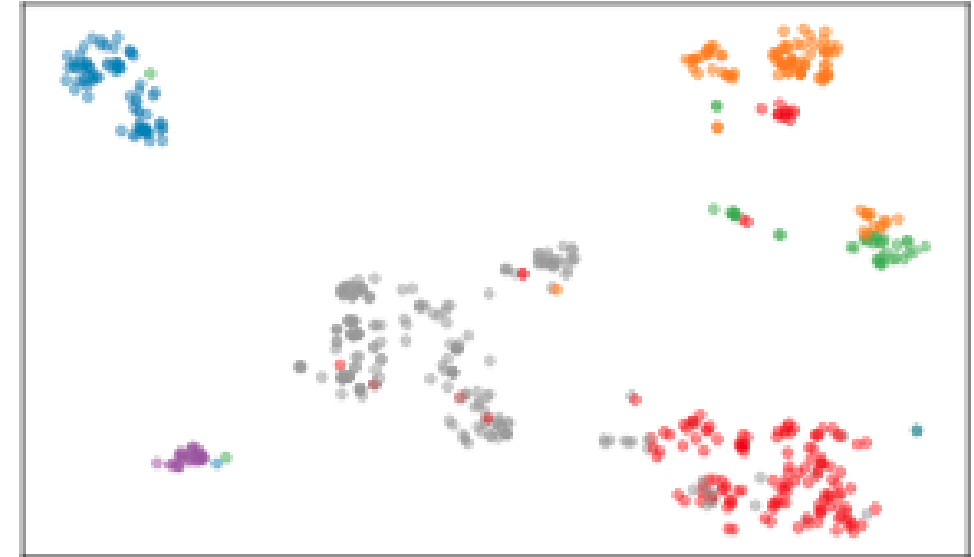
Our Results



With-label: qualitative results



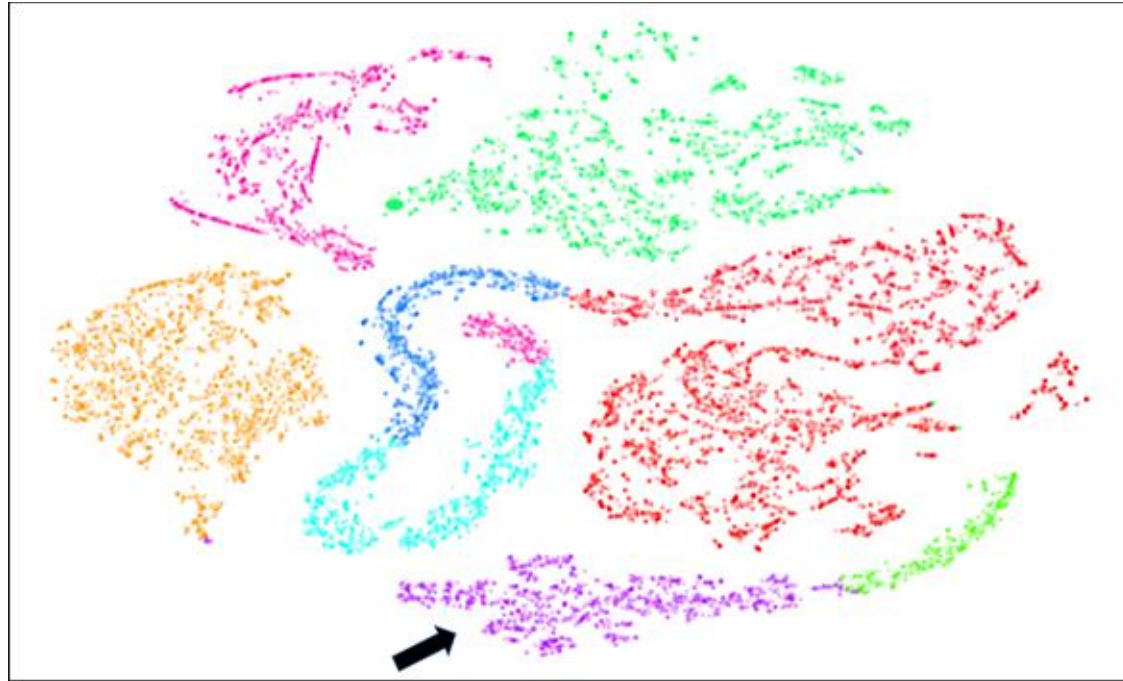
Embedding after Phase I



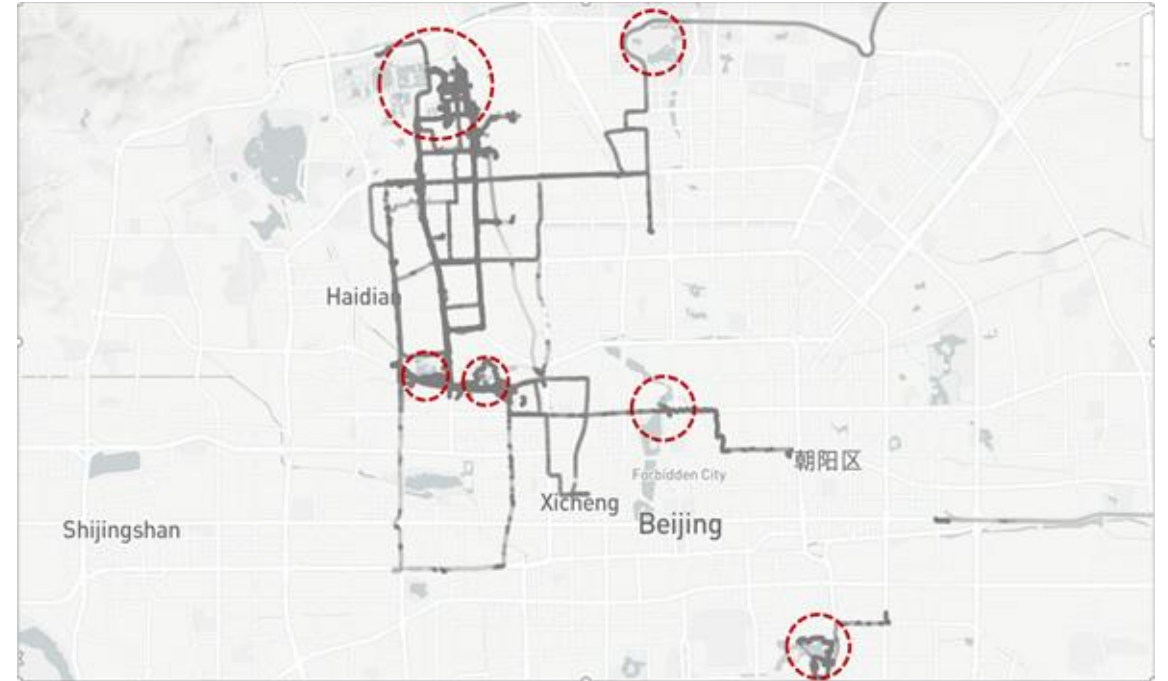
Embedding after Phase II



Without-label: qualitative results



Embedding of full dataset



Recreation Activities

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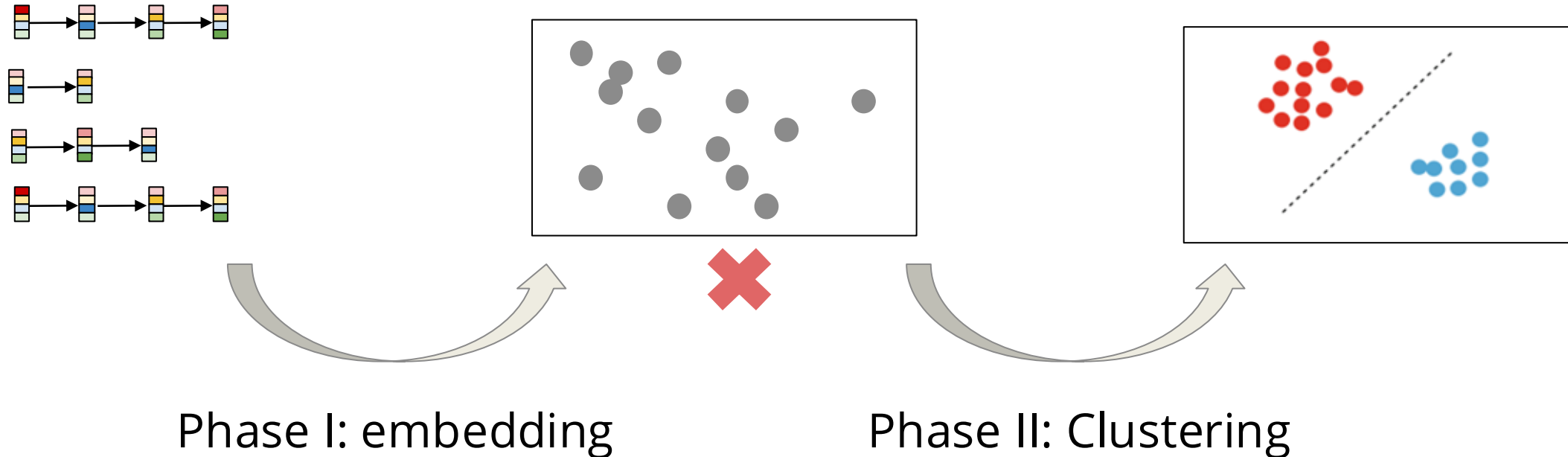


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Shortcoming: Separate Embedding from Clustering



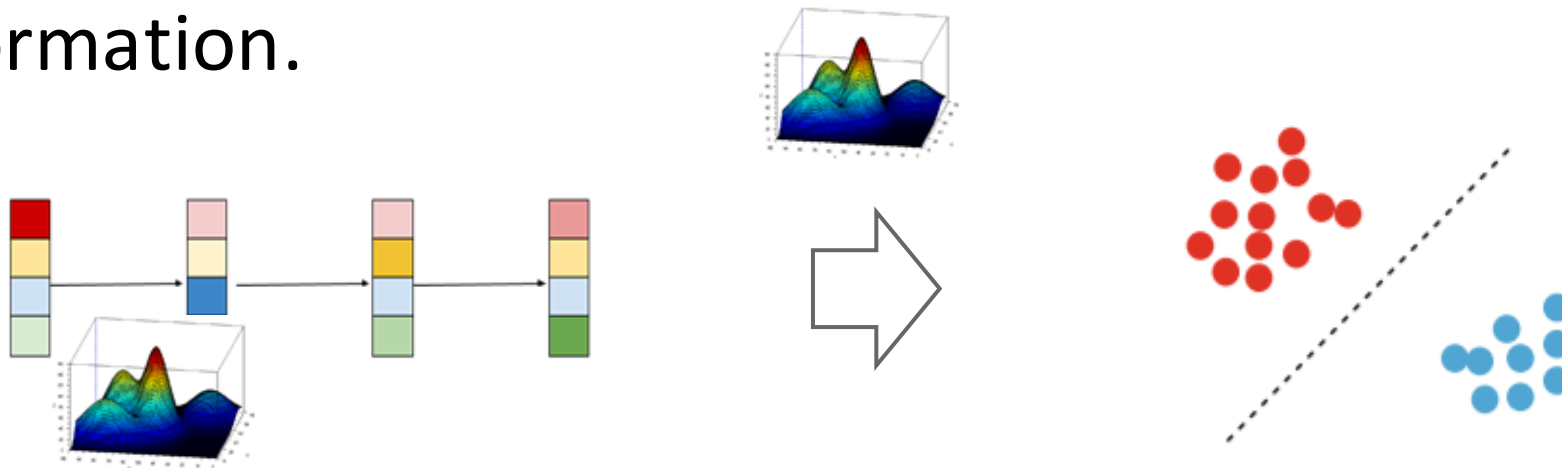
Two-phase: sub-optimal since the first phase is not aware of clustering.





Solution: A single phase

Assume the pre-existence of clusters in the latent space and jointly learn the hidden representation for reconstruction and cluster formation.

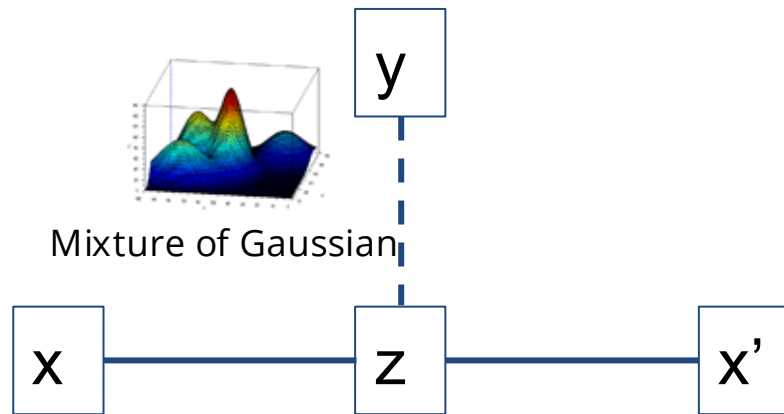




VAE: Variational AutoEncoders

VAE-based (1 phase) for images VaDE [IJCAI 17], GMVAE [ArXiv 16], JointVAE [NIPS 18]

$$y \sim \text{Cat}(1/K), z \sim \mathcal{N}(\mu_y, \sigma_y^2 I)$$
$$x \sim \mathcal{N}(\mu_x(z), \sigma_x^2(z) I)$$



y : discrete(onehot) variable indicating cluster

z : continuous variable denoting embedding

x : input context sequences

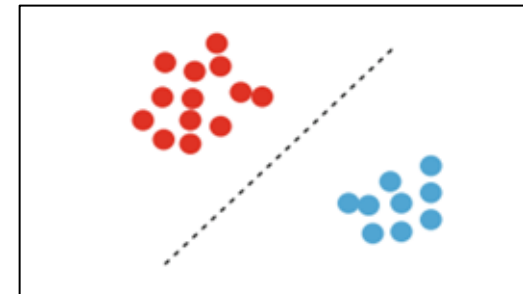
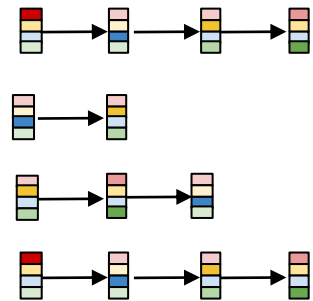
x' : reconstructed context sequences

Challenge: Sensitive Training of Clustering



Two-phase: sensitive since the first phase is not aware of clustering.

One-phase: produce trivial solutions as the model could ignore the cluster involvement.



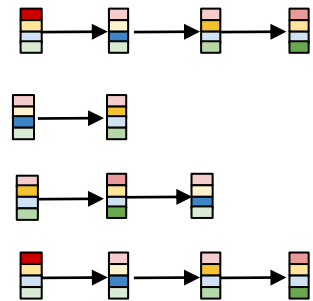
Embedding + Clustering

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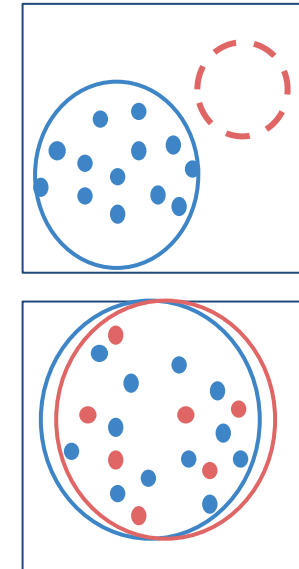


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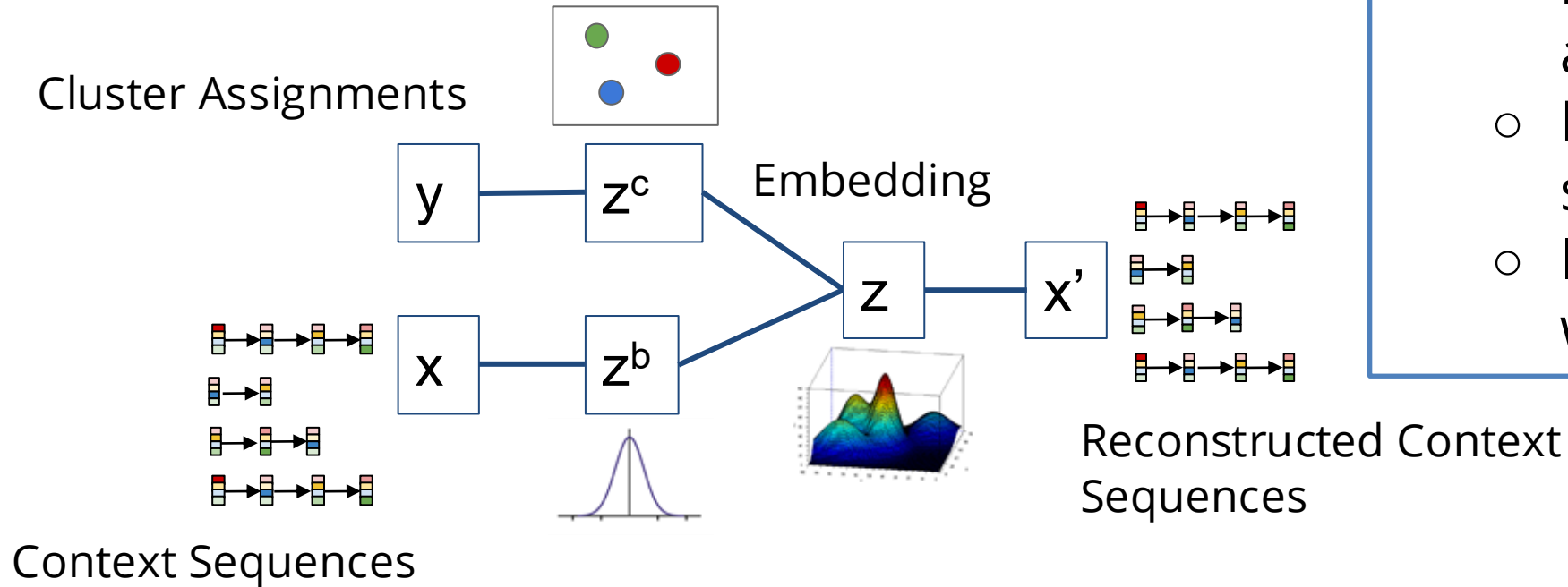
One-phase: produce trivial solutions as the model could ignore the cluster involvement.



Embedding + Clustering



VAMBC: A Variational Approach for Mobility Behavior Clustering



- Increase the involvement of cluster assignments y .
- Encourage the cluster separation
- Improve the cohesion within each cluster

VAMBC





VAMBC

$$\begin{aligned}\mathcal{L}_{ELBO} &= \mathbb{E}_{y, z \sim q(y, z|x)} \left(\log \frac{p(y)}{q(y|x)} + \log \frac{p(z^b)}{q(z^b|x)} + \log p(x|y, z) \right) \\ &= -D_{KL}(q(y|x) || p(y)) - D_{KL}(q(z^b|x) || p(z^b)) \\ &\quad + \mathbb{E}_{y \sim q(y|x), z^b \sim q(z^b|x), z^c = f(y; W)} \log p(x|y, z^b, z^c)\end{aligned}$$



VAMBC

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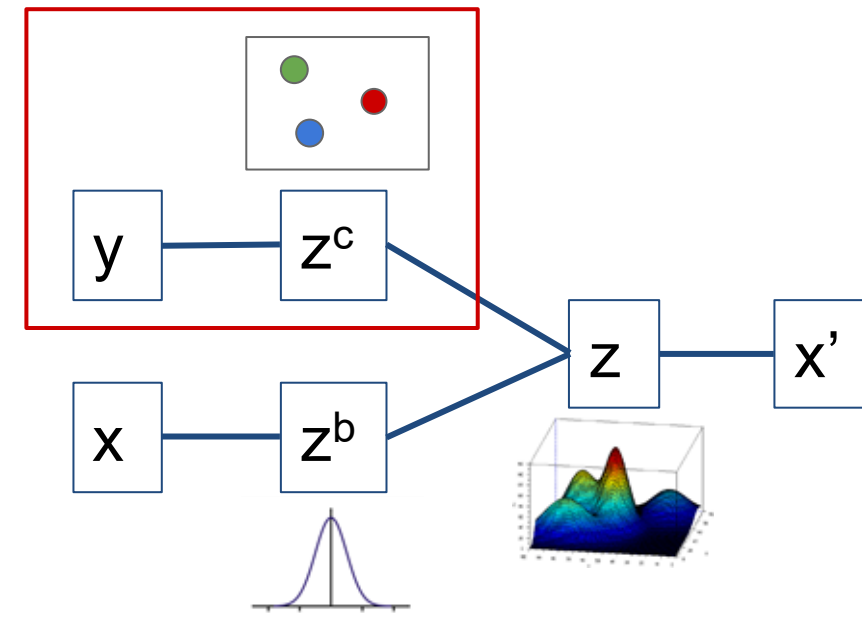
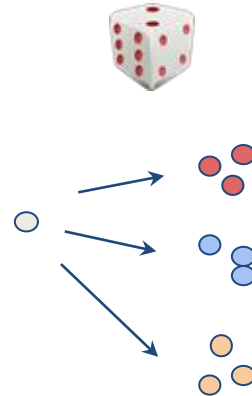
VAMBC

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$\mathcal{L}_{NE}$: Negative Entropy

- Bring randomness to y
- Allow cluster adjustment
- Share z^c

Gumbel-Softmax



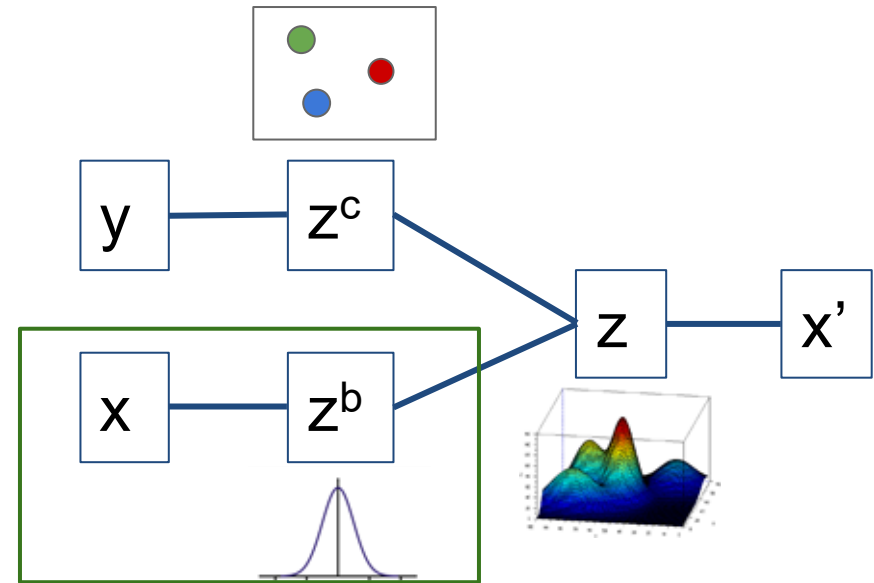
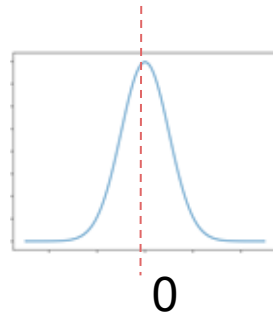


VAMBC

$$\begin{aligned}\mathcal{L}_{ELBO} &= \mathbb{E}_{y, z \sim q(y, z|x)} \left(\underbrace{\log \frac{p(y)}{q(y|x)}}_{\mathbf{L}_{NE}} + \log \frac{p(z^b)}{q(z^b|x)} \underbrace{+ \log p(x|y, z)}_{\mathbf{L}_{KL}} \right) \\ &= - \underbrace{D_{KL}(q(y|x) || p(y))}_{\text{red box}} - \underbrace{D_{KL}(q(z^b|x) || p(z^b))}_{\text{green box}} \\ &\quad + \underbrace{\mathbb{E}_{y \sim q(y|x), z^b \sim q(z^b|x), z^c = f(y; W)} \log p(x|y, z^b, z^c)}_{\mathbf{L}_{recon}}\end{aligned}$$

$\mathbf{L}_{KL}$: KL loss

- A regularizer limits z^b .
- z^b only carries unique information of x .



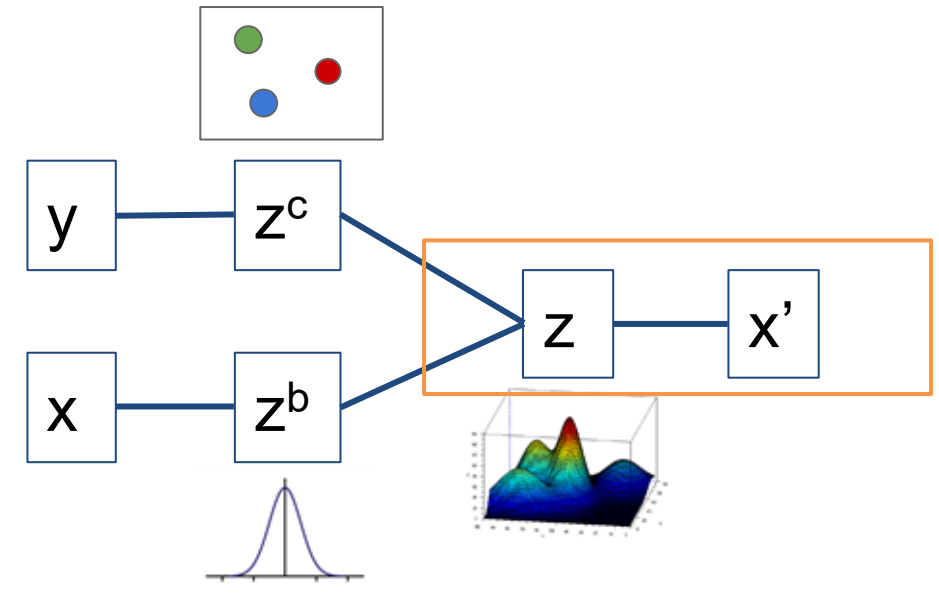
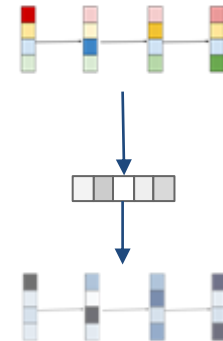


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$\mathbf{L}_{recon}$: Reconstruction

- Supervise the entire model
- learn the embedding z





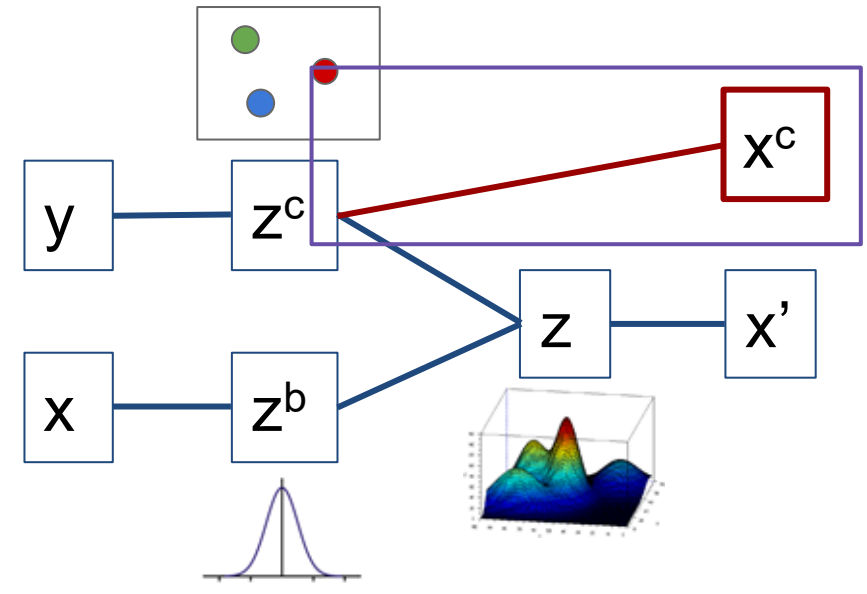
VAMBC

$$L = L_{NE} + L_{KL} + L_{recon} + L_{center}$$

L_{center} : Center loss

- Force z^c meaningful
- More involvement of y

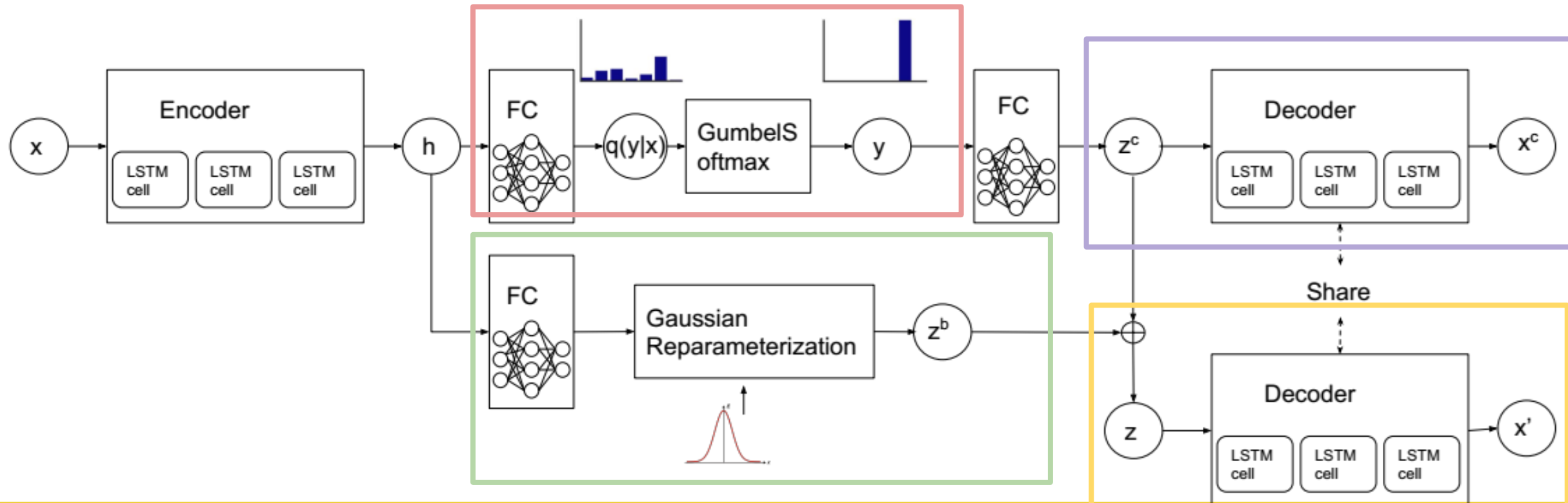
$$L_{center} = ||x - x^c||_2^2$$





VAMBC Architecture

$$\text{LOSS} = L_{\text{NE}} + L_{\text{KL}} + L_{\text{recon}} + L_{\text{center}}$$



Experiment: setting



- **Dataset:**
 - GeoLife trajectories and POI data
 - DMCL trajectories and POI data
- **Baselines:**
 - Classical TS: KM-DTW, KM-GAK, kShape, DB-LCSS
 - Autoencoder-based: DTC, DETECT, IDEC*, DCN*
 - VAE-based: VaDE*, GMVAE*, JointVAE*
 - Discrete Sequence Clustering: SGT, MHMM
- **Metrics:**
 - NMI: Normalized Mutual Information
 - ARI: Adjusted Rand Index
 - Accuracy

Experiment: result



Method	NMI (aver)	NMI (best)	NMI (worst)	ARI(aver)	ARI (best)	ARI (worst)	ACC(aver)	ACC (best)	ACC (worst)
KM-DTW	0.610±0.021	0.645	0.579	0.635±0.019	0.656	0.617	0.742±0.031	0.763	0.655
KM-GAK	0.591±0.057	0.657	0.507	0.505±0.076	0.573	0.392	0.737±0.033	0.770	0.688
K-Shape	0.229±0.033	0.272	0.174	0.220±0.046	0.271	0.102	0.522±0.015	0.551	0.495
DB-LCSS	0.547	0.547	0.547	0.412	0.412	0.412	0.697	0.697	0.697
SGT	0.419±0.024	0.454	0.371	0.216±0.036	0.277	0.149	0.628±0.029	0.694	0.579
MHMM	0.530±0.047	0.611	0.486	0.403±0.057	0.495	0.344	0.627±0.017	0.649	0.607
IDEC*	0.605±0.035	0.673	0.572	0.465±0.097	0.664	0.404	0.67±0.08	0.819	0.596
DCN*	0.646±0.051	0.725	0.594	0.635±0.065	0.693	0.503	0.782±0.061	0.840	0.624
DTC	0.500±0.027	0.550	0.474	0.483±0.028	0.512	0.451	0.682±0.032	0.737	0.655
DETECT	0.644±0.037	0.691	0.589	0.646±0.044	0.688	0.582	0.8±0.013	0.822	0.780
GMVAE*	0.132±0.267	0.501	0.250	0.098±0.128	0.351	0.000	0.411±0.079	0.551	0.346
VaDE*	0.631±0.053	0.669	0.502	0.603±0.078	0.658	0.440	0.783±0.037	0.822	0.720
JointVAE*	0.459±0.056	0.556	0.408	0.227±0.123	0.442	0.161	0.519±0.062	0.597	0.473
VAMBC	0.697±0.015	0.699	0.692	0.7±0.019	0.719	0.682	0.825±0.01	0.842	0.810

Conclusion



- Proposed a novel variational model VAMBC that clusters the context sequences in a single phase
- The VAMBC model decomposes the hidden embedding into individualized embedding and cluster embedding with a novel design.
- The VAMBC model achieves significantly better robustness and improved accuracy than existing approaches.

OUTLINE



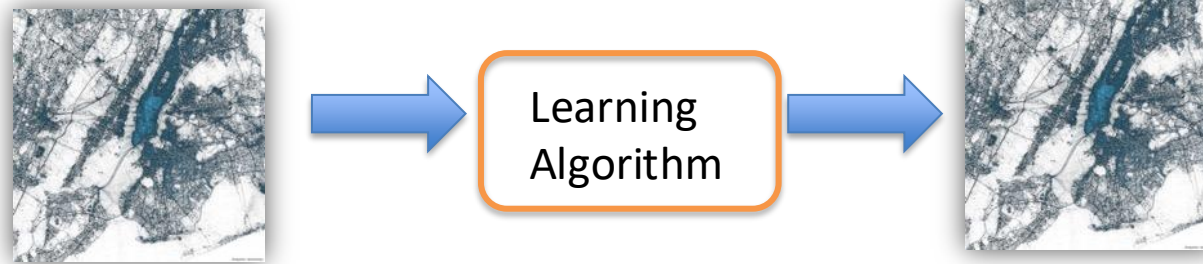
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 - DETECT
 - VAMBC
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OUTLINE



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Synthetic Trajectory Generation – Why?



Real Data

Generated Data

- **Scale-up:** The input set is small and we need more realistic trajectories for the downstream task or when
- **Privacy preservation:** The privacy of the input set must be preserved
- **Diversification:** To generate trajectories for one geographical area (e.g., city, neighborhood) from the training set of trajectories belonging to a different geographical area

Background



Computational algorithms

- Moving-object generators (e.g., Brinkhoff)
 - Travel surveys and Handcraft-rules
- Micro agent simulators
 - Lots of parameter setting

Limitation: Require human labor and domain knowledge to convert real-world data into the parameters/rules of the closed (artificial) environment.

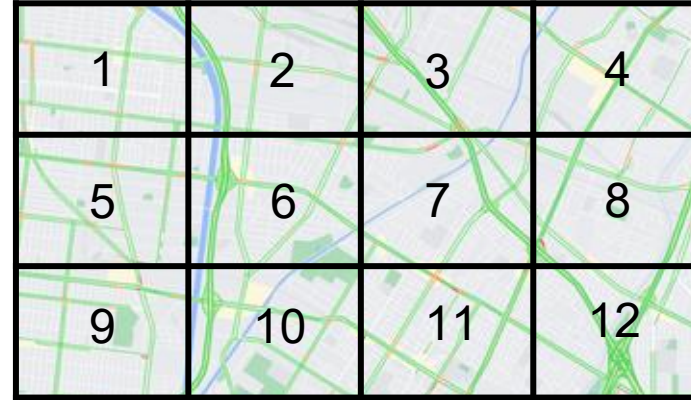
Data-Driven algorithms

- Learn directly from the Real data distribution

Discretize



Latitude, longitude, timestamp



Mobility Trajectory

[1 , 3 , 7, 8 , 8]

8:00, 8:05, 8:10, 8:15, 8:20...

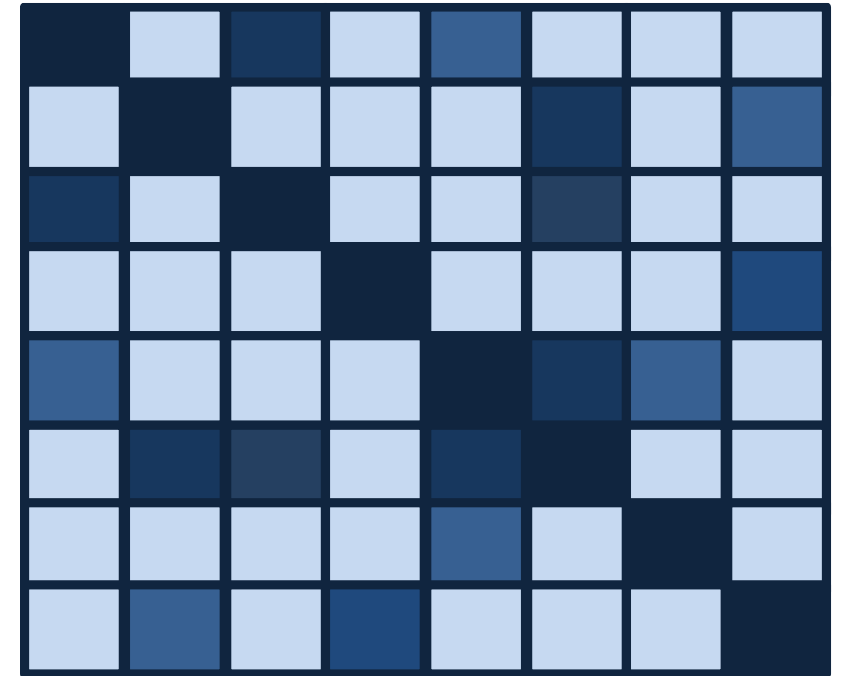
- Discrete Trajectory in space and time



Generative AI Idea

Model human movements as state transitions and formulate trajectory generations as a decision-making process

Follow a generator-discriminator structure to learn the underlying mechanism (e.g., policy) behind the state transitions.

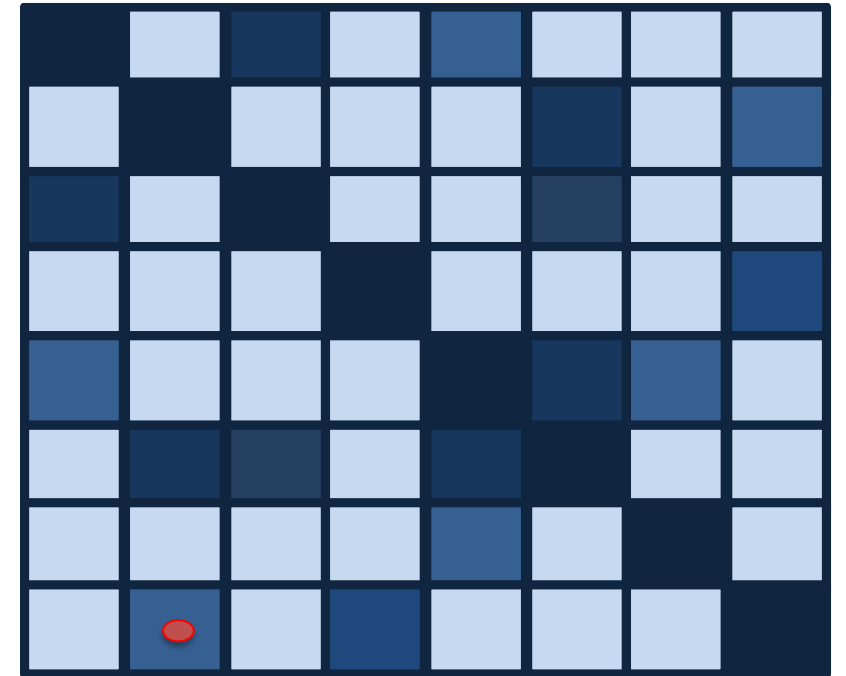




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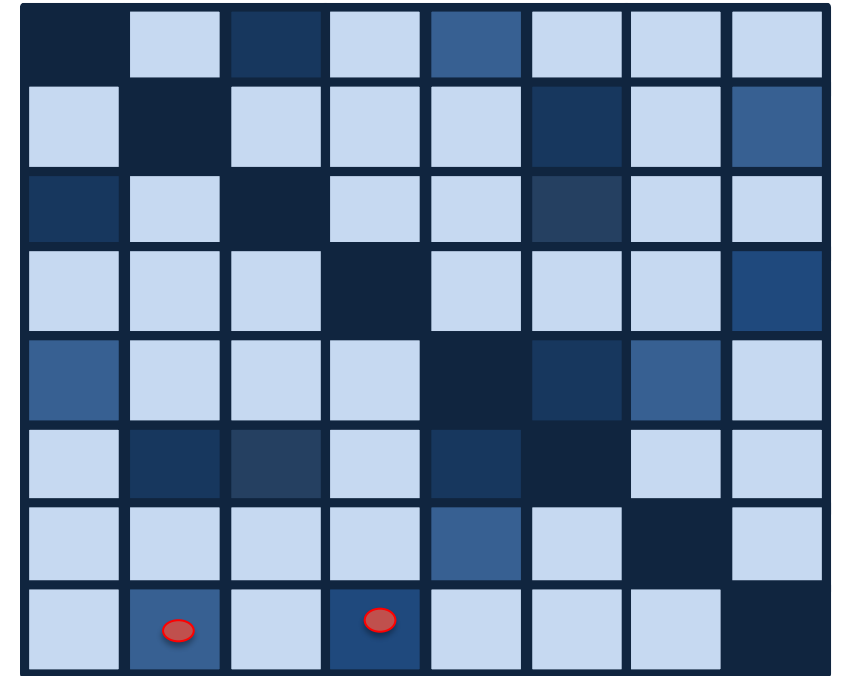




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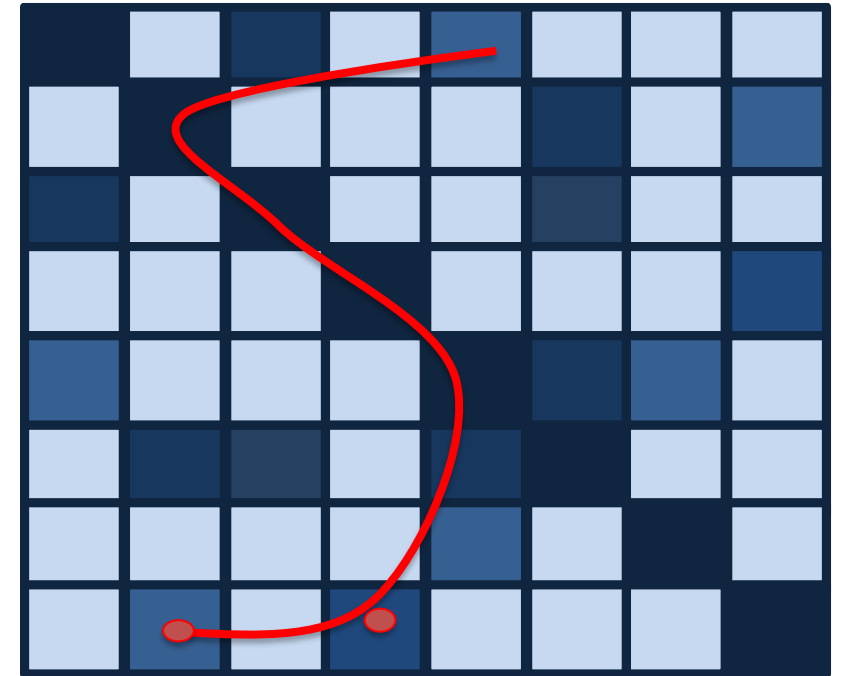




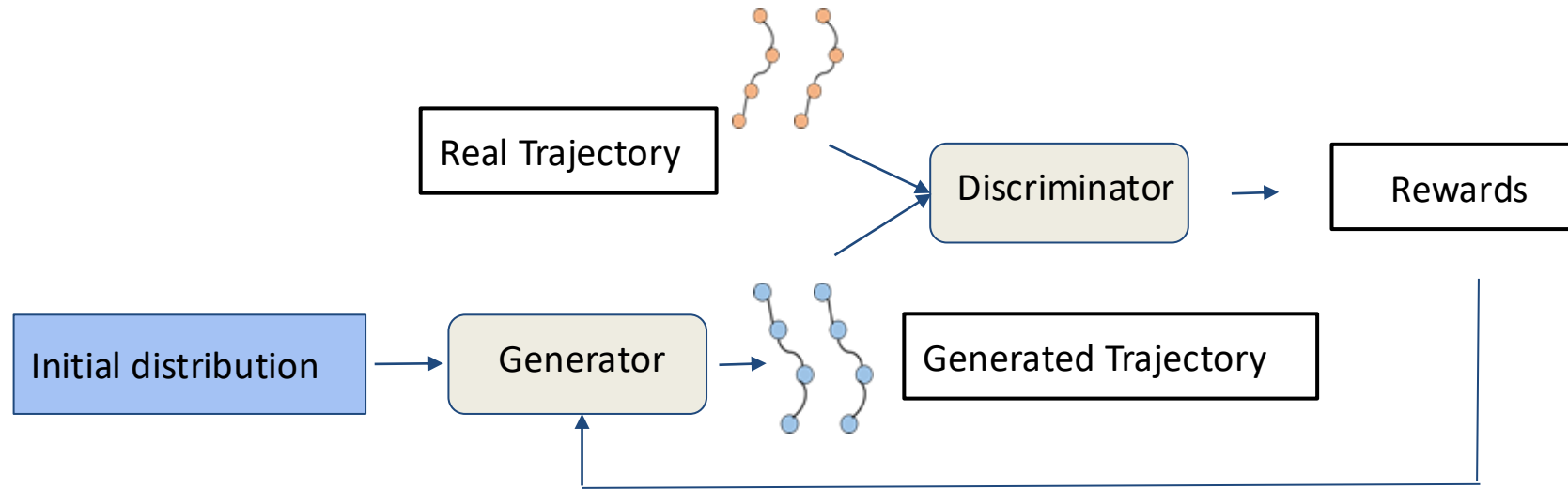
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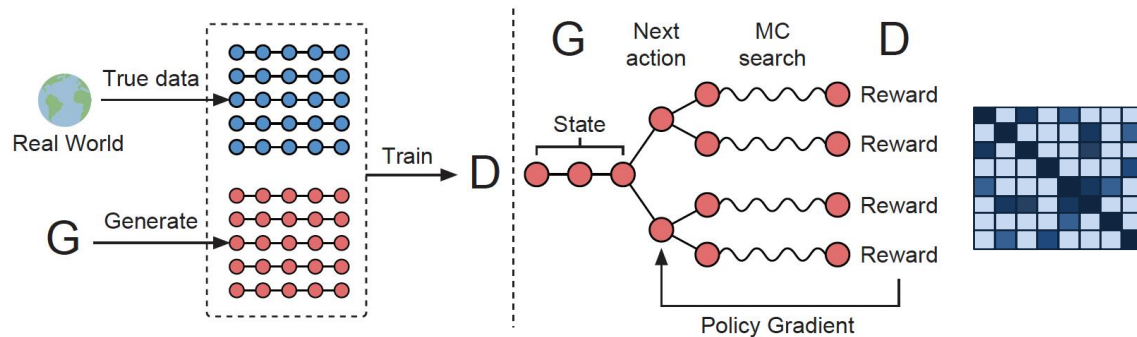
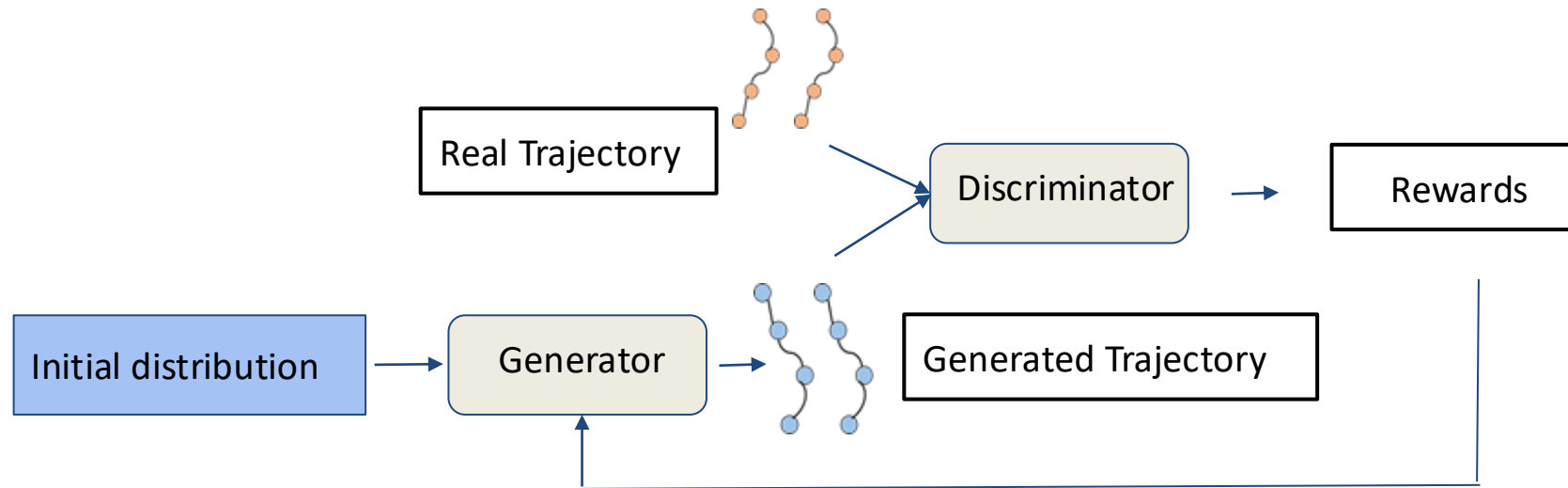
Background: Seq Generative Adversarial Network



$$\min_{G_\theta} \max_{D_\phi} E_{Y \sim p_d(Y)} [\log(D_\phi(Y))] + E_{\hat{Y} \sim G_\theta(\hat{Y})} [\log(1 - D_\phi(\hat{Y}))]$$

Compete with each other in a zero-sum game framework.

Background: Seq Generative Adversarial Network



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Related Work



	Name	Methods	Dataset
Non Deep Learning Model	TimeGeo [PNAS16]	Statistics modeling	CDR(Boston)
DNN Methods	TrajGAN [IJCAI18]	GAN	Nokia Lausanne location
	MoveSim [KDD 20]	GAN	GeoLife (Beijing)
	TrajGAIL [ICDM20]	GAIL	Taxi data (Shenzhen)
	DeltaGAN [ICLR 21]	GAN	GeoLife (Beijing)
	NEXTGAIL [SIGSPATIAL21]	GAIL	Taxi data (Shenzhen)
	ActSTD [KDD 21]	GAIL	Mobile Network (Beijing) Foursquare
	FVAE [SIGSPATIAL22]	VAE	Taxi data (Porto) Gowalla

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Our Main Hypothesis

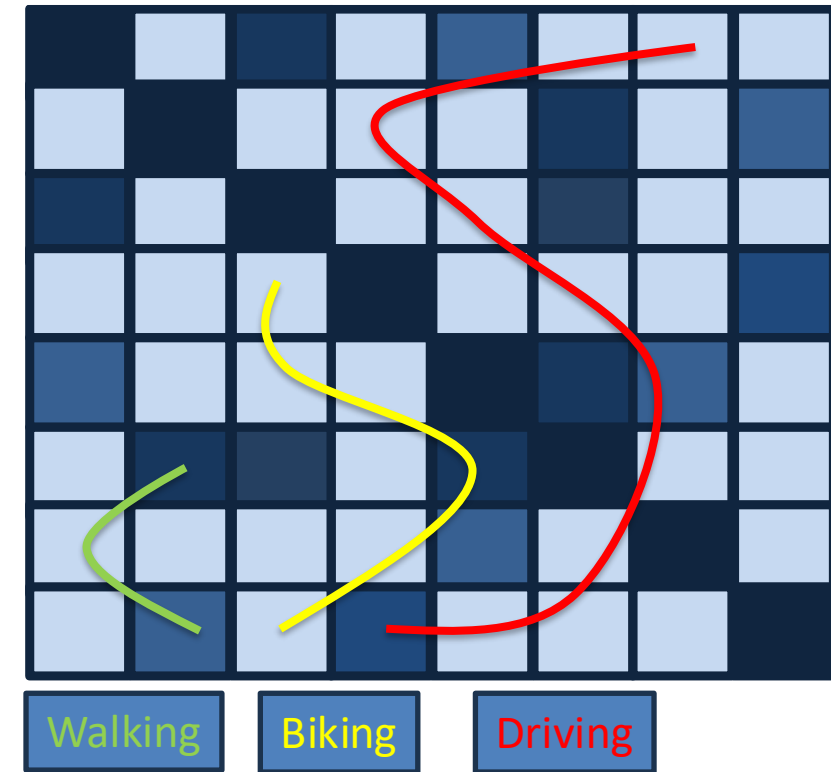


While common mobility regularity and transition patterns are shared across transportation modalities and moving behaviors, there are also modality/behavior-specific characteristics and patterns.

E.g., different speeds, distances traveled, number of distinct visits, and transition patterns (e.g., transitions on walkways vs. bikeways vs. roads).

By including modality/behavior, the generated trajectories will be

*Diverse and representative of different modalities
Realistic, corresponding to the real-world modalities*



OUTLINE



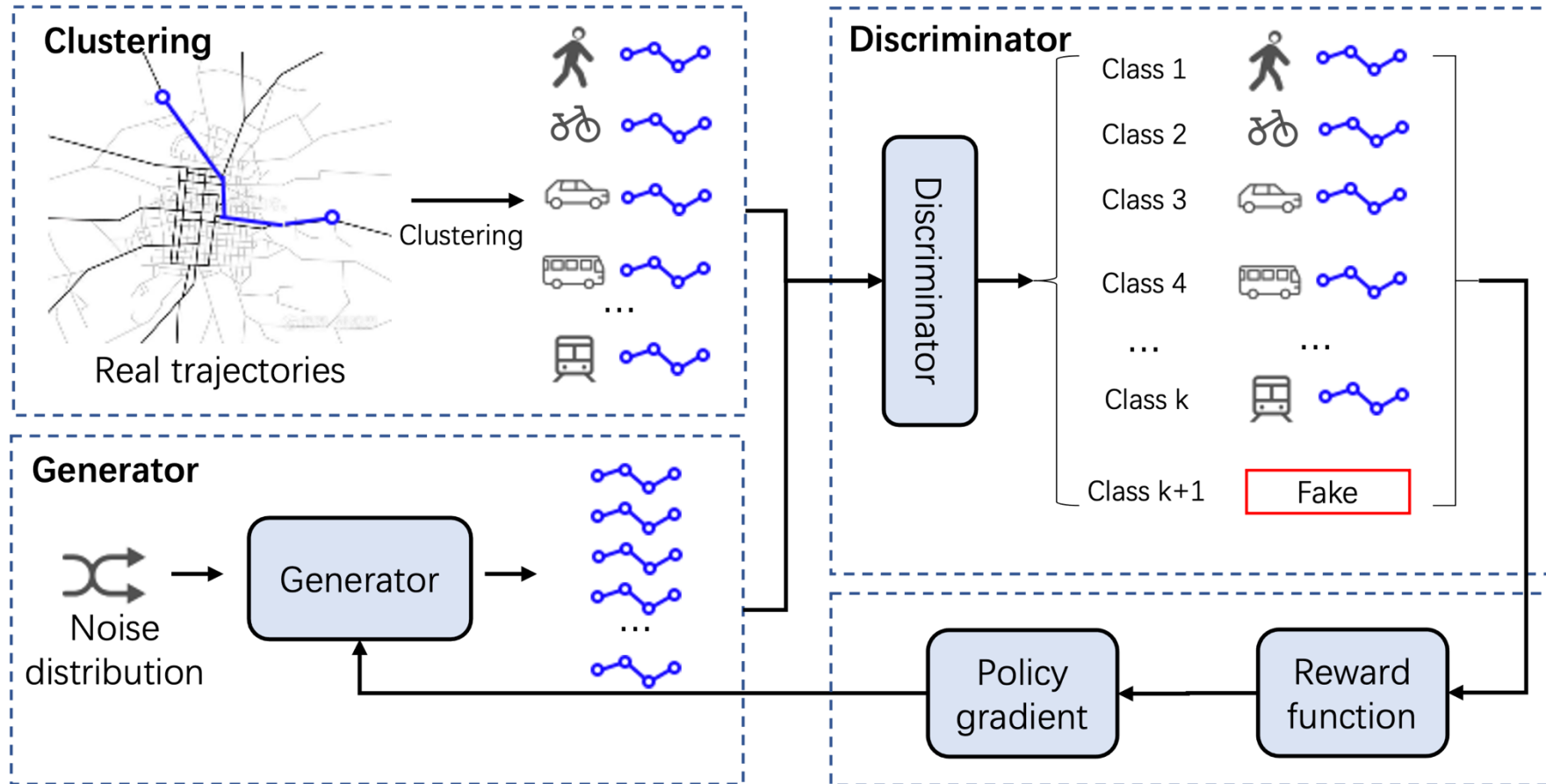
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CSGAN: Clustering-based Sequence Generative Adversarial Network



- Classify the trajectory into $k + 1$ classes
- Rewards of the generator is the summation of any real modalities

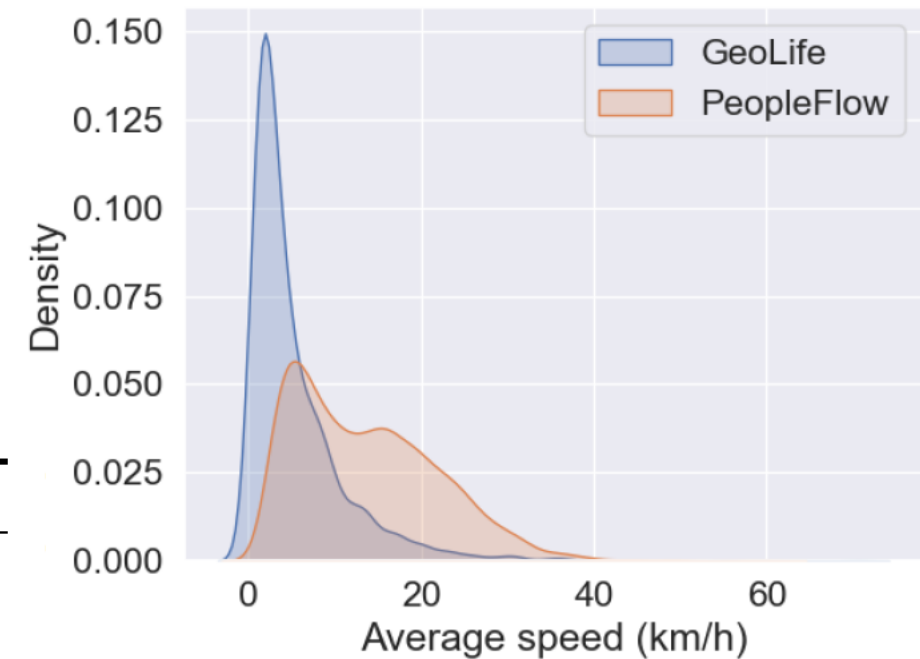
$$R_D(\hat{Y}) = \sum_{c \in C_r} p_D^c(\hat{Y})$$

Datasets



- GeoLife
 - Collected by Microsoft Research Asia from 182 users
- PeopleFlow
 - Tokyo Metropolitan Area
 - Transportation mode for each visit is available

Characteristics	GeoLife	PeopleFlow
Number of Trajectories	2756	6183
Period	1 year (2008)	1 year (2008)
Visit Interval	every 15 minutes	every 15 minutes
Average Speed (km/h)	5.324 ± 5.744	13.592 ± 8.303
Average Accumulated Distance (km)	8.401 ± 9.340	27.662 ± 26.494
Average Distinct Visits	7.713 ± 4.558	7.123 ± 3.895



PeopleFlow has more diverse speeds due to mix-modalities



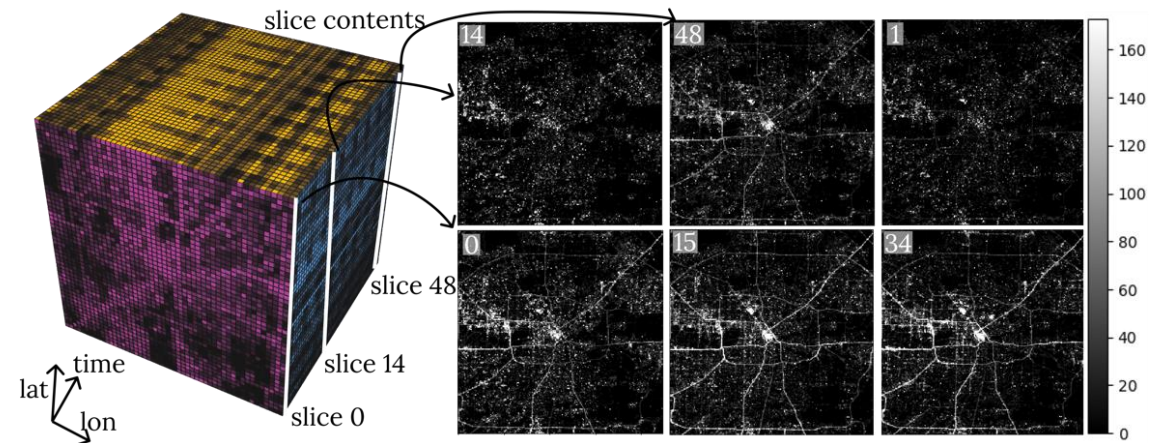
Evaluation Metrics

Geographical density-based statistics

Probability of visiting location
r, probability of visiting
location r at time t

Individual trajectory-level statistics

Distance traveled $P(d)$,
number of distinct visits $P(v)$



$$JSD(P_{re}||P_{gen}) = H\left(\frac{P_{re} + P_{gen}}{2}\right) - \frac{H(P_{re}) + H(P_{gen})}{2}$$

Jensen-Shannon Divergence (JSD) between the probability distribution of the real trajectories and the generated trajectories for each distribution.



Our Newly Proposed Evaluation Metrics

Transition statistics

Probability of transitioning from location r_1 to r_2

Modality level statistics

PC0: The proportion of trajectories within the correct cluster (clustering based on centroids of real trajectories)

PC1: Clustering separately (for both real and generated trajectories) and performing centroid matching and JSD of the two distributions is reported.

C: Given the vector of k centroids from real trajectories and generated trajectories, we compute the minimum total pair-wise distance among all permutations (closest match). The lower the value, the better the generated trajectories preserve the modality representatives.

Frobenius norm of the difference between the two transition matrices

$$\|P_r - P_g\|_F = \sqrt{\sum_{r_1=1}^Q \sum_{r_2=1}^Q |P_r(r_1, r_2) - P_g(r_1, r_2)|^2}$$

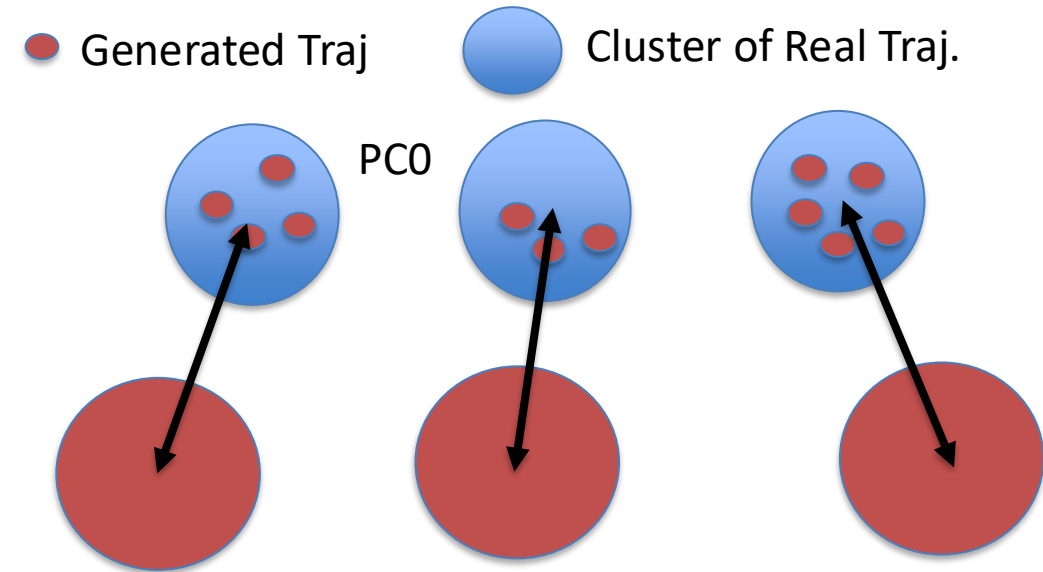




TABLE II: Global comparison with baselines on GeoLife and PeopleFlow data with different clustering techniques. The table shows the average statistics of 5 experiments. The best performance is in boldface. The second-best is underlined.

Methods	GeoLife							
	Geographical density-based statistics		Individual trajectory level statistics		Transition statistics	Modality level statistics		
	P(r)	P(r,t)	P(d)	P(v)	P(r1, r2)	$P(c_i^0)$	$P(c_i^1)$	C
SeqGAN-S	<u>0.407</u>	<u>0.478</u>	<u>0.208</u>	0.288	0.100	<u>0.162</u>	0.220	18.682
Cluster-wise SeqGAN-S	<u>0.506</u>	<u>0.562</u>	<u>0.313</u>	0.234	<u>0.082</u>	0.402	0.298	83.441
Movesim-S	0.522	0.579	0.263	0.390	0.116	0.303	<u>0.209</u>	7.934
CSGAN-S	0.319	0.439	0.195	<u>0.258</u>	0.047	0.147	0.058	<u>17.128</u>

PeopleFlow								
SeqGAN-S	0.378	0.437	0.368	<u>0.275</u>	<u>0.092</u>	0.406	0.167	29.694
Cluster-wise SeqGAN-S	<u>0.344</u>	0.406	<u>0.363</u>	0.317	0.105	0.311	0.194	31.046
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SeqGAN on
each cluster

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Cluster SeqGAN: Not enough data per cluster to learn global mobility patterns. It outperforms SeqGAN with PeopleFlow bc the modality is more diverse (vs GeoLife) and hence it is more important to learn modality-specific patterns within each cluster.



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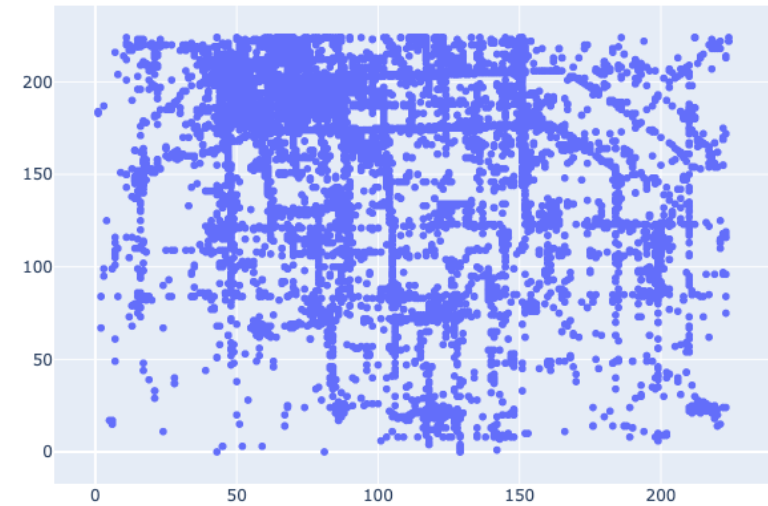
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CSGAN-S	0.284	0.376	0.218	0.215	0.050	0.146	0.070	13.136

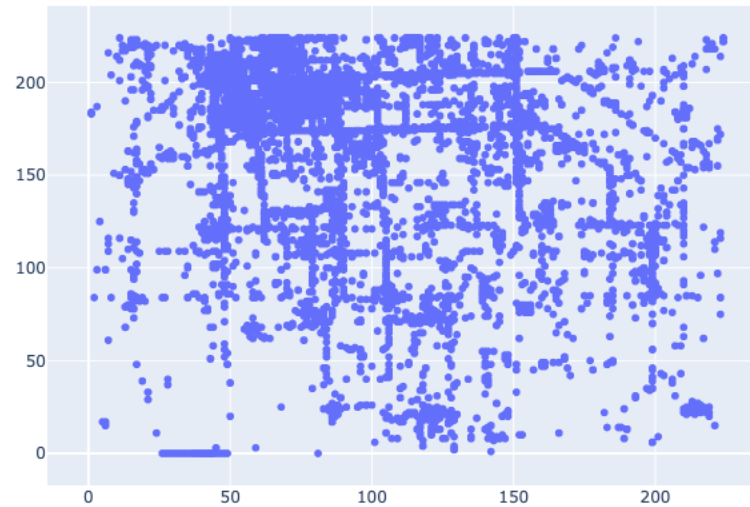
Cluster SeqGAN: Not enough data per cluster to learn global mobility patterns. It outperforms SeqGAN with PeopleFlow bc the modality is more diverse (vs GeoLife) and hence it is more important to learn modality-specific patterns within each cluster.

Discriminator's
Spatial & temporal
regularities check

Real (GeoLife)



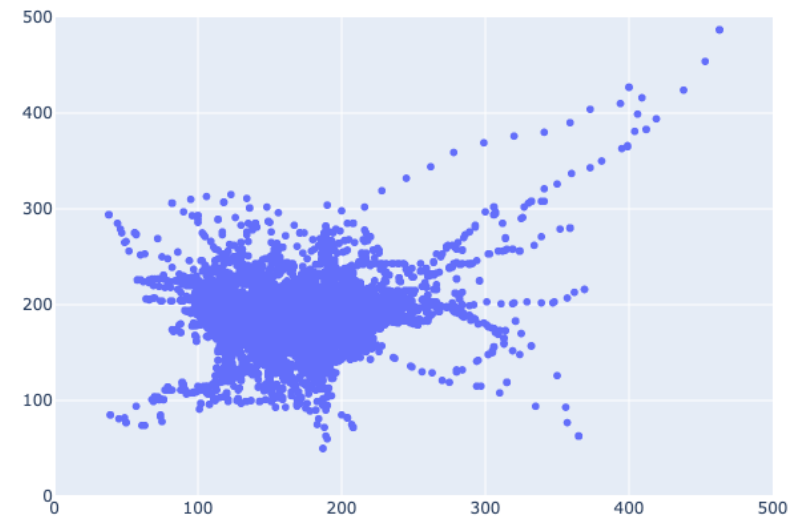
CSGAN (GeoLife)



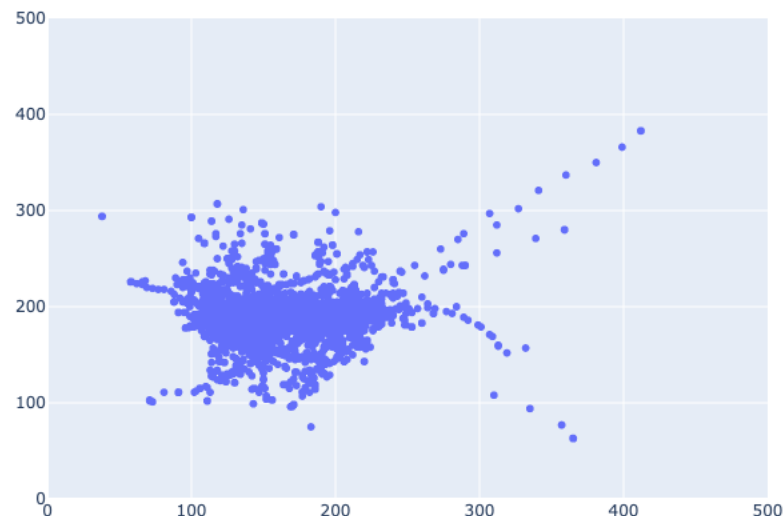
Density Visualization



*The population density --
The aggregate density from
6:00 am to 8:00 pm*



Real (PeopleFlow)



CSGAN (PeopleFlow)

OUTLINE



- Mobility Behavior Clustering
 - DETECT
 - VAMBC
- Synthetic Trajectory Generation
 - CSGAN
 - MPGAIL

OUTLINE



- Mobility Behavior Clustering
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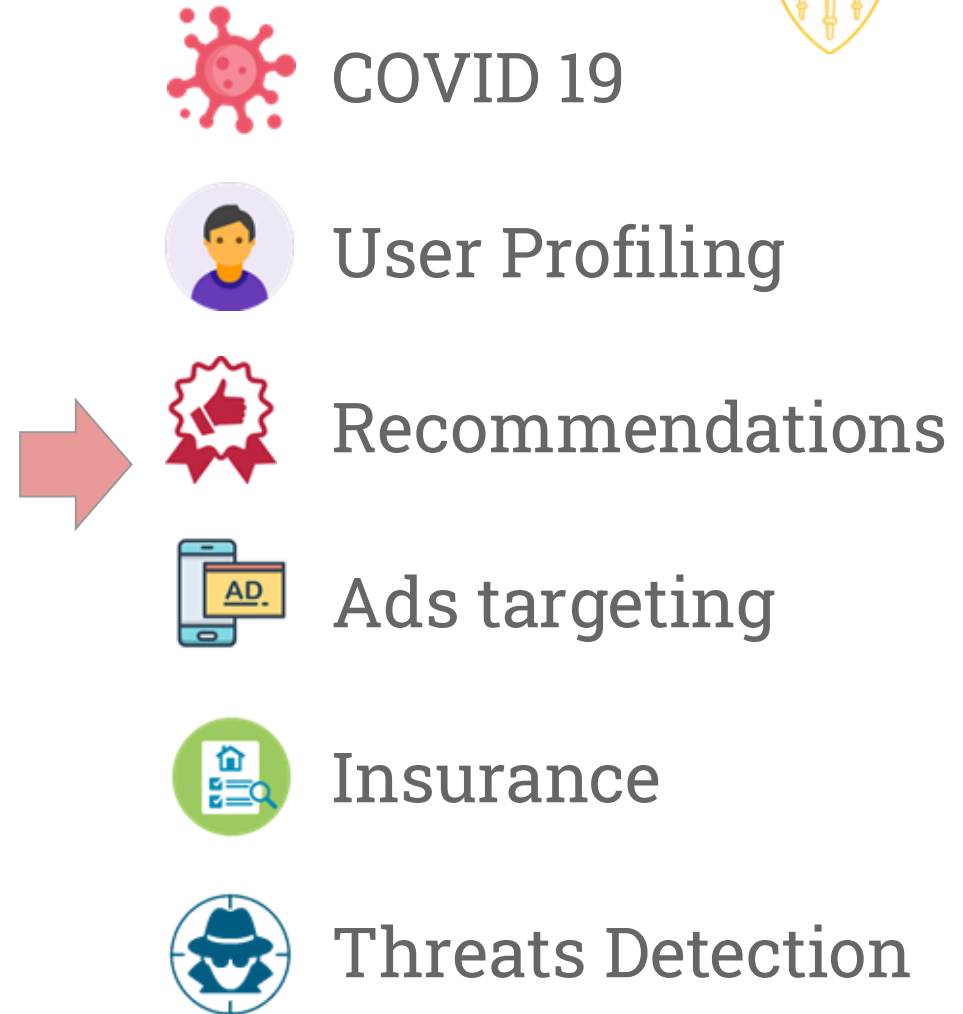
Motivation



Mobility behavior:
the travel activity that describes a user's movements, e.g., work commute, shopping, school commute, dining



Mobility Behavior



Context Trajectory

- Discretizing mobility trajectory by grid/time Partition



Mobility Trajectory

[1 , 3 , 7 , 8 , 8]

Context Trajectory

- Discretizing mobility trajectory by grid/time Partition



Latitude, longitude, timestamp



Context Trajectory

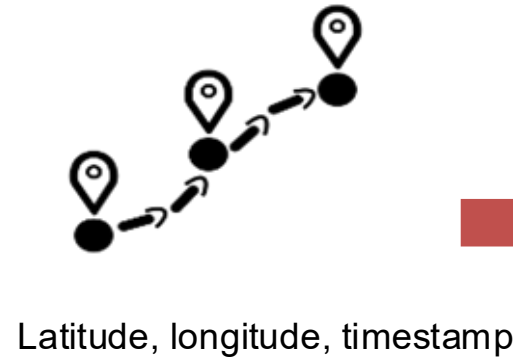
Mobility Trajectory



[1 , 3 , 7 , 8 , 8]

Context Trajectory

- Discretizing mobility trajectory by grid/time Partition

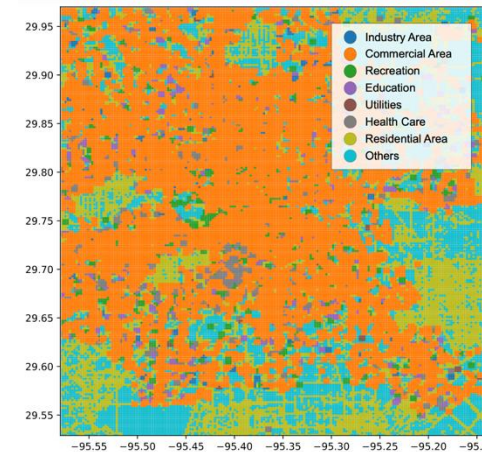


Context Trajectory

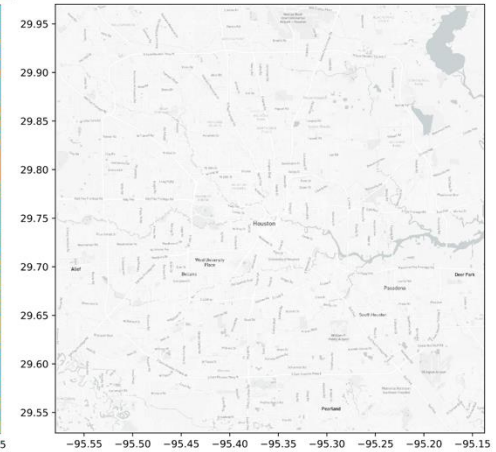
- Augment location with context
 - Decide by the number of Point of Interests (POIs)
 - Industry, Commercial Area, Entertainment, Education Services, Utilities, Health Care, Residential Areas, & Others
- Similar Moving Behavior has similar context



[1 , 3 , 7 , 8 , 8]



(a) The Context Type of Each Location Grid in Houston



(b) The Raw Map of Houston

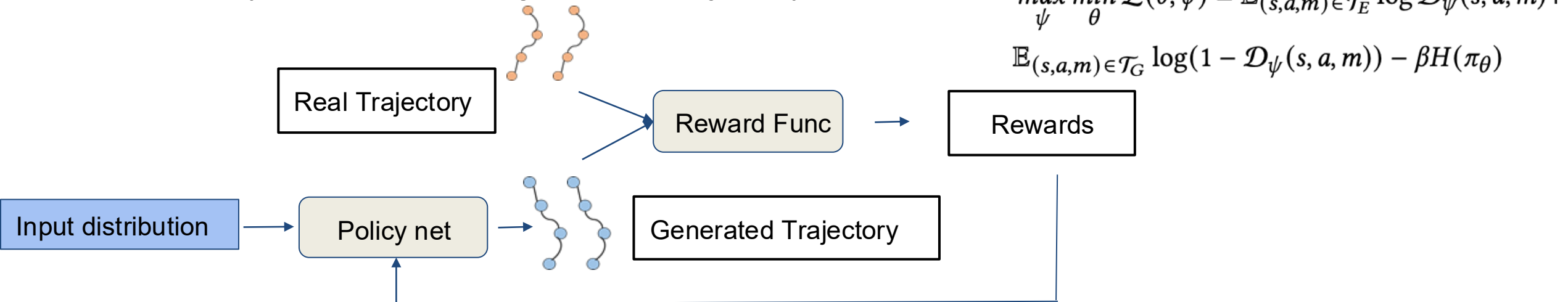


Preliminaries & Overview

Problem : Given a set of real-world trajectories and their moving behavior labels, how to generate synthetic trajectories while retaining moving behavior properties

GAIL : Generative Adversarial Imitation Learning

- Goal : modeling human agent decision-making process from real-data
 - policy: the underlying strategy to generate the trajectory
 - reward: ways to evaluate the generated trajectory





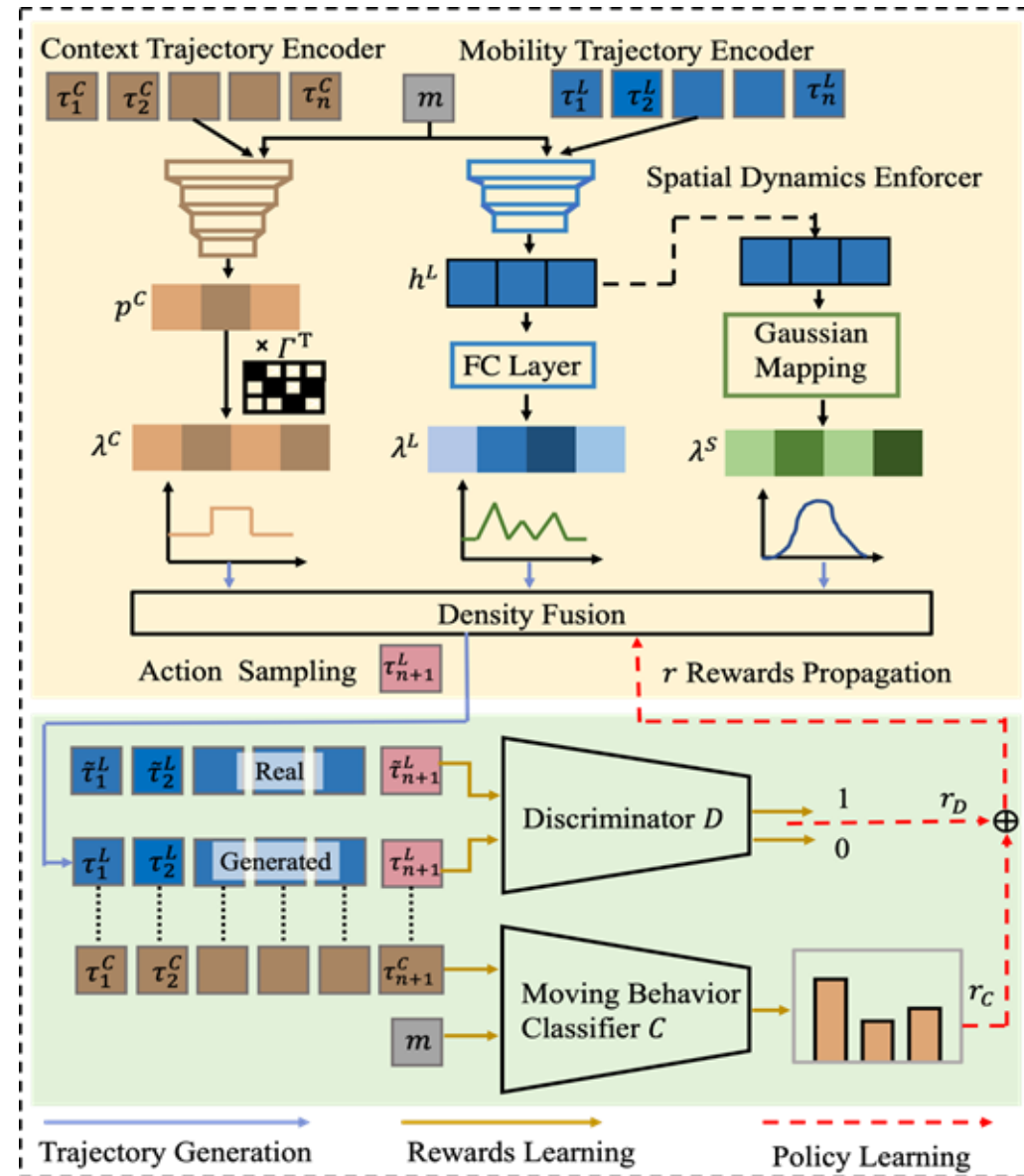
Preliminaries & Overview

Problem : Given a set of real-world trajectories and their moving behavior labels, how to generate synthetic trajectories while retaining moving behavior properties

- State: the history of the generated trajectory until the last step
- Action: which location to go next
- State transition: update the mobility trajectory with the chosen location
- Policy network: characterizes how to choose the following action, given the moving behavior
- Reward: evaluate the decision of chosen action based on the current state



MBP-GAIL Framework



MBP-GAIL Framework

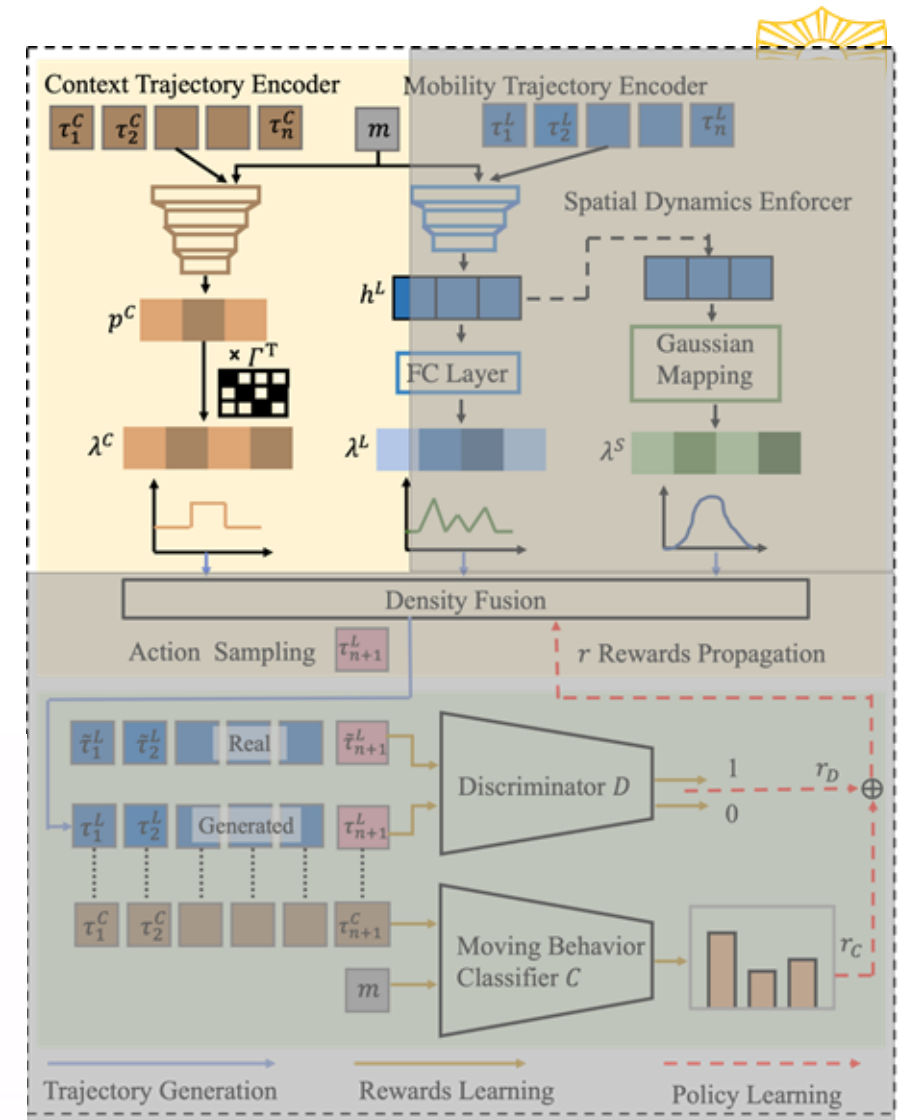
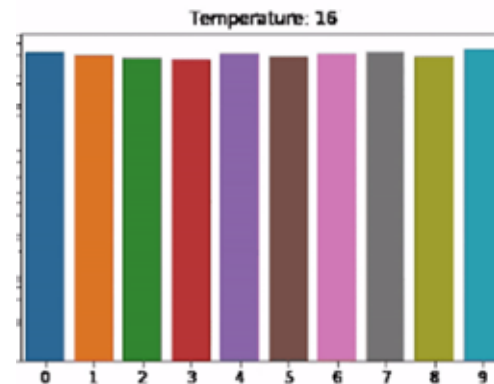
Context Trajectory Encoder:

$$p_i^C = \text{Softmax}(\phi(W_h^C \times h_i^C + b_h^C), \gamma)$$

predict the next context type based on the
history with temperature γ
Map to each location

$$\lambda_i^C = \Gamma^T \times p_i^C$$

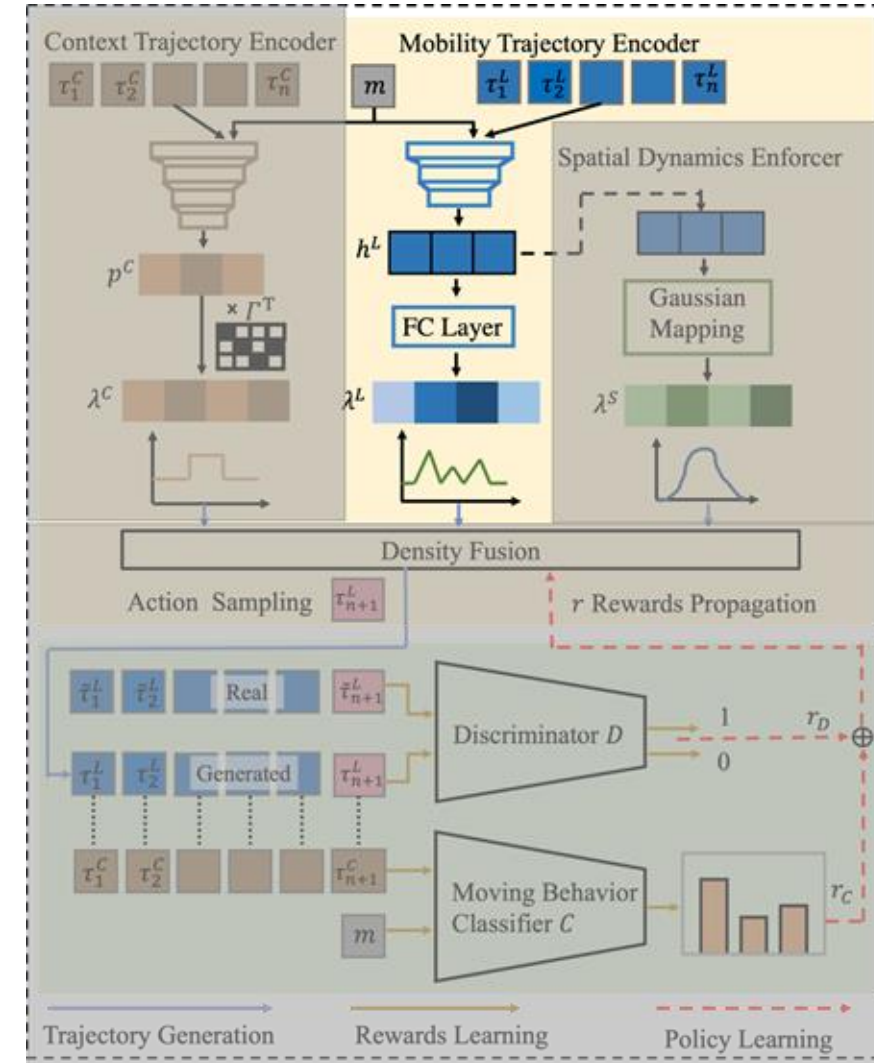
As we set the temperature higher,
the context has less influence on our
final generation





MBP-GAIL Framework

Mobility Trajectory Encoder:
predict the next location based on the
history





MBP-GAIL Framework

Spatial dynamic enforcer $\lambda_i^S = \exp(-\frac{(\Delta \times \text{OneHot}(l_i))^2}{(\alpha \sigma_i^S)^2})$

trajectory should be physically feasible

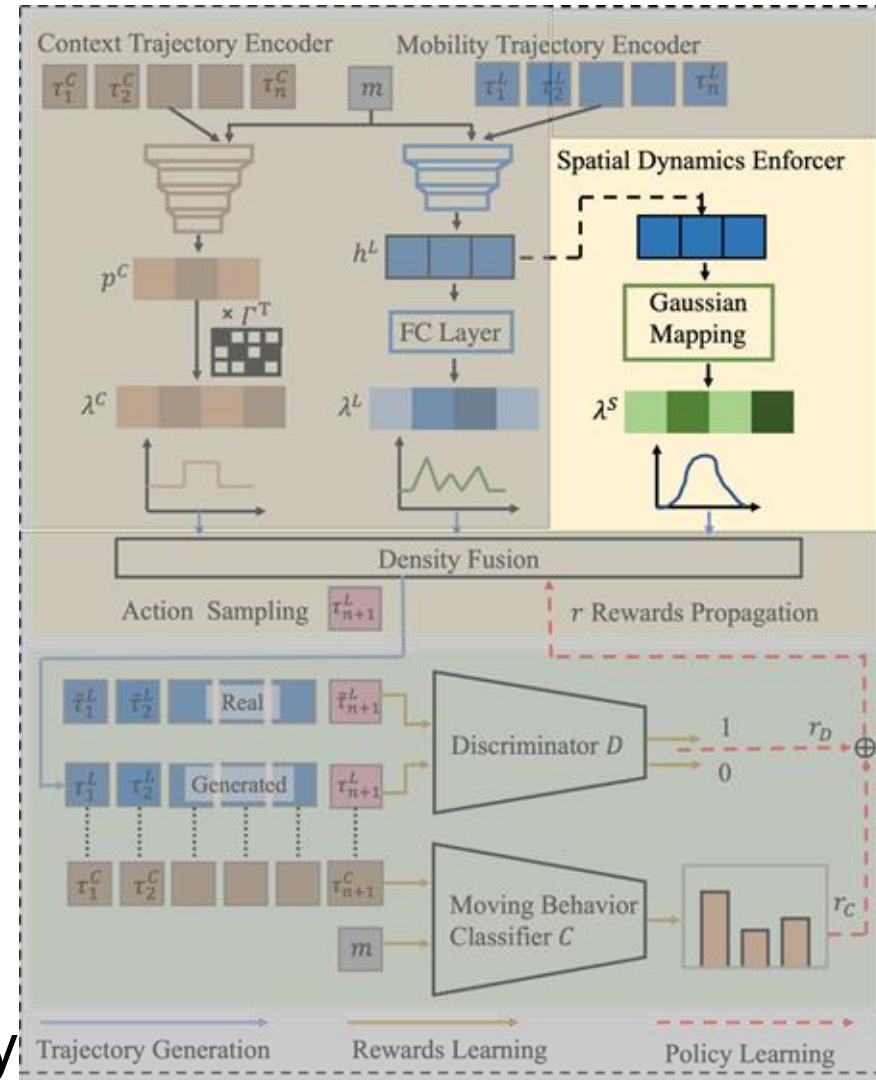
Higher probability



Lower probability



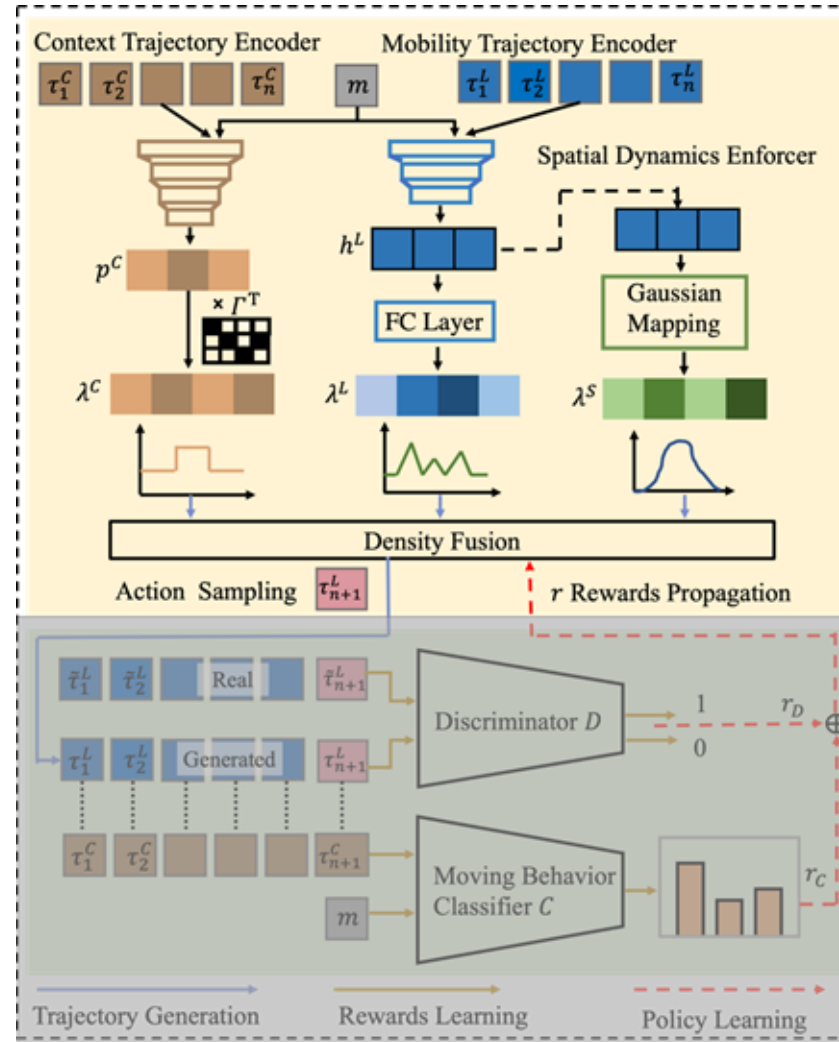
fixed constraint due to variety transportation mode





MBP-GAIL Framework

Fusion
element wise multiplication





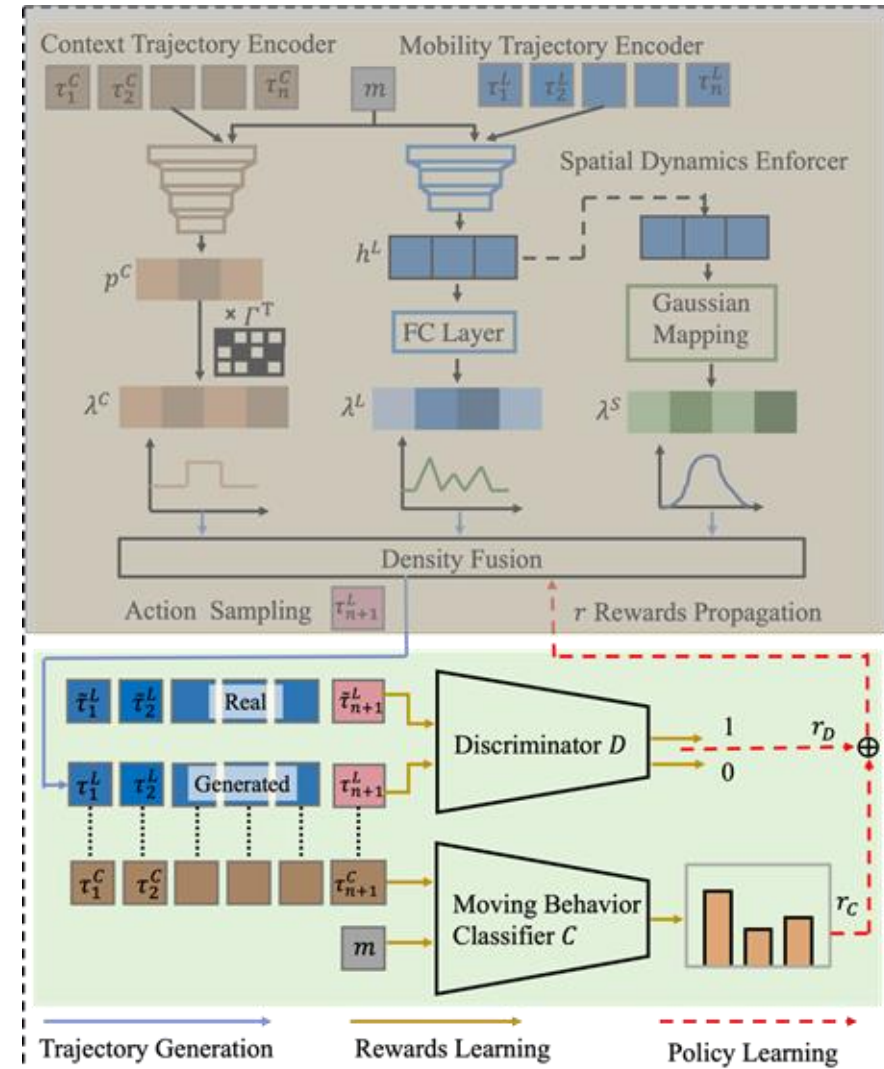
MBP-GAIL Framework

a discriminator differentiates
real/fake trajectory
a classifier evaluates moving
behavior pattern

$$r = (1 - v) * r_C + v * r_D$$

Reward from
moving
behavior

Reward from
discriminator





Experimental Evaluation

	Houston					Los Angeles				
	Distance	Radius	Duration	$P(r)$	$P(r_1, r_2)$	Distance	Radius	Duration	$P(r)$	$P(r_1, r_2)$
Markov Model	0.5098	0.5032	0.4428	0.0028	0.3280	0.4086	0.4122	0.4332	0.0046	0.3073
LSTM	0.4865	0.4050	0.3748	0.0023	0.0881	0.3855	0.3050	0.3830	0.0032	0.1044
TransVAE	0.4662	0.3942	0.3276	0.0034	0.1537	0.3872	0.3443	0.3539	0.0042	0.1462
SeqGAN	0.3318	0.2908	0.2160	0.0074	0.1055	0.2948	0.1913	<u>0.1490</u>	0.0025	<u>0.0910</u>
MoveSim	<u>0.2413</u>	<u>0.2402</u>	<u>0.1520</u>	0.0025	<u>0.0924</u>	<u>0.0922</u>	0.1274	0.1617	0.0021	0.0932
MBP-GAIL	0.0744	0.1215	0.1311	<u>0.0024</u>	0.0874	0.0667	<u>0.1305</u>	0.1452	<u>0.0023</u>	0.0891

- For realistic evaluation (utility compared with original trajectory), MBP-GAIL outperforms almost all the JSD metrics evaluation, especially for the distance over the best baseline.



Experimental Evaluation

Much better in distance-related metrics
due to spatial dynamic enforcer

	Distance	Houston				Los Angeles				
		Radius	Duration	$P(r)$	$P(r_1, r_2)$	Distance	Radius	Duration	$P(r)$	$P(r_1, r_2)$
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- For realistic evaluation (utility compared with original trajectory), MBP-GAIL outperforms almost all the JSD metrics evaluation, especially for the distance over the best baseline.



Experimental Evaluation

Much better in distance-related metrics due to spatial dynamic enforcer

Much better transition due to context clustering modeling

	Houston					Los Angeles				
	Distance	Radius	Duration	$P(r)$	$P(r_1, r_2)$	Distance	Radius	Duration	$P(r)$	$P(r_1, r_2)$
Markov Model	0.5098	0.5032	0.4428	0.0028	0.3280	0.4086	0.4122	0.4332	0.0046	0.3073
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- For realistic evaluation (utility compared with original trajectory), MBP-GAIL outperforms almost all the JSD metrics evaluation, especially for the distance over the best baseline.



Experimental Evaluation

- **RQ2:** Can MBP-GAIL preserve the moving behavior patterns in its generation?
 - Compare with movesim / seqgan which also knows the moving behavior information
 - Clustering on the generated context trajectories
 - Closer to the real-world data distribution, the better



U MBP-GAIL preserves the mobility trajectory patterns and achieves the lowest gap compared with real-data



Thank you!